

The Future of Growth and Human Labor Under Recursive AI Self-Improvement

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Abstract

I develop an endogenous-growth model in which a monopolistic AI developer supplies AI services and invests in recursive AI self-improvement. I characterize two long-run equilibrium regimes under an upper bound on AI efficiency. Complementarity between AI and labor or a sufficiently low upper bound can leave labor as a production bottleneck. In this regime, improvements in AI efficiency can temporarily raise growth, but the growth rates of output per worker and wages eventually return to the exogenously given rate of labor-augmenting technological progress. When AI and labor are substitutes and the upper bound is sufficiently high, the economy sustains faster wage growth and a higher long-run return to capital than in the labor-bottleneck regime. However, output per worker grows even faster than wages, so labor's income share converges to zero while AI-industry revenue rises as a share of output. Numerical simulations show that output-per-worker growth may remain moderate for a long time before accelerating in the AI-dominated regime or returning to the rate of labor-augmenting technological progress in the labor-bottleneck regime.

[Digital edition and interactive simulations](#)

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1 Introduction

Advances in artificial intelligence raise both hopes and fears. One hope is that increasingly capable AI could generate unprecedented economic prosperity, with workers benefiting from higher wages. One fear is that the AI industry could capture a growing share of that prosperity while workers who depend on labor income receive a declining share. In this paper, I show how both outcomes can coexist: when AI and labor are substitutes, AI can permanently raise wage growth. Yet in the long run, output per worker grows even faster than wages, while AI-industry revenue grows faster than total labor income. Labor’s income share therefore converges to zero.

To show this, I extend the Ramsey–Cass–Koopmans model to include AI services as an additional production input. Households choose saving, and competitive firms produce the final good using capital, effective labor, and AI services. A monopolistic AI developer supplies these services and invests in recursive AI self-improvement (RSI) to improve AI efficiency. As in [Romer \(1990\)](#), expected monopoly profits provide incentives for research investment. By introducing a monopolistic AI developer, I can distinguish how the gains from AI are distributed through wages, returns to physical capital, and AI-industry profits.

The model combines two feedback loops related to those in [Davidson et al. \(2026\)](#) and [Jones and Tonetti \(2026\)](#). A technological loop operates through RSI: greater AI efficiency produces more AI research services from a given amount of research compute and thereby helps create still better AI. An economic loop runs from AI efficiency to output and back to research: greater AI efficiency raises production and expands the resources available to fund research compute. With substitution and sufficiently strong research returns, the developer’s discounted net profit can become unbounded if AI efficiency is unrestricted. To rule this out, I first study an economy with an upper bound on AI efficiency and examine how equilibria change as the bound rises.

Together, the elasticity of substitution between AI production services and labor and the maximum attainable level of AI efficiency determine whether human labor remains a long-run production bottleneck or whether AI accelerates economic growth. I focus on two long-run equilibrium regimes, distinguished by whether AI produces a temporary or permanent increase in growth. In the labor-bottleneck regime, AI improvements can raise growth temporarily, but human labor ultimately constrains production. In the AI-dominated regime, capital and AI services sustain permanently faster growth.

I first characterize equilibria in which labor remains a long-run production bottleneck. Under complementarity, I establish their existence when the upper bound on AI efficiency exceeds an elasticity-dependent threshold. At unit elasticity, any upper bound suffices. With substitution, such equilibria exist when the upper bound lies below the corresponding threshold. Along these equilibria, output per worker and wages eventually grow at the exogenous rate of labor-augmenting technological progress, and labor’s income share converges to a positive value.

The outcome changes when the elasticity of substitution between AI production services and labor exceeds one and the maximum attainable level of AI efficiency lies above the threshold. An AI-dominated long-run regime then becomes possible. As in [Jones and Manuelli \(1990\)](#), capital

accumulation can sustain growth when its limiting return remains sufficiently high. In this regime, capital and effective AI production services eventually grow faster than effective labor. This does not require AI efficiency to keep improving: capital accumulation and expanding compute use sustain growth even as AI efficiency approaches its upper bound. The long-run growth rates of output per worker and the wage, as well as the interest rate, exceed their labor-bottleneck values. Because output per person grows faster than wages, labor’s income share converges to zero. Workers therefore receive higher wages, but these wages do not keep pace with capital income not the AI-industry revenue.

I also examine the effect of removing the upper bound on AI efficiency. With complementarity, long-run output-per-worker growth cannot exceed the exogenous rate of labor-augmenting technological progress. At unit elasticity, I derive conditions for a balanced-growth equilibrium in which output per worker grows faster than this exogenous rate. Wages grow at the same rate as output per worker, preserving a constant labor income share. With substitution and sufficiently strong research returns, the developer can make discounted net profit arbitrarily large even over a fixed finite horizon. In such a case there is no equilibrium.

I simulate an unanticipated activation of RSI in economies that already use AI services. Depending on research productivity, growth can display a moderate increase or a noticeable jump no matter the regime. Eventually, growth returns to the rate of labor-augmenting technological progress in the labor-bottleneck regime or accelerates even further in the AI dominated regime. If research productivity is low enough the initial trajectories between regimes look pretty similar.

This paper has seven sections, including the present introduction. Section 2 relates the paper to the literature. Section 3 presents the model and defines equilibrium. Section 4 establishes sufficient conditions for equilibrium with an upper bound on AI efficiency and characterizes the associated long-run growth and distributional outcomes. Section 5 examines the three substitution regimes without an upper bound on AI efficiency. Section 6 presents numerical simulations. Section 7 concludes. The appendix contains the proofs and describes the algorithm used for the simulations.

2 Related literature

This paper addresses the connected questions of growth, income distribution, and transition dynamics in the research agenda proposed by [Brynjolfsson et al. \(2025\)](#). My closest antecedents are models in which AI improves both production and research. [Davidson et al. \(2026\)](#) and [Jones and Tonetti \(2026\)](#) study feedback between technological progress, research, and output. Technological progress is endogenous in both papers, but saving and research allocation shares are taken as given. Another close antecedent is [Erdil et al. \(2025\)](#), who model how investment in compute, training, and technology expands both the range of tasks AI can perform and the services it supplies within that range. Whereas they solve a social planner’s problem, I study a market equilibrium with a monopolistic AI developer. I likewise distinguish resources used to deploy AI from those used to improve it. I focus on how RSI increases the services obtained per unit of compute, without

explicitly modeling the automation of additional tasks.

The developer’s research incentives build on [Romer \(1990\)](#), where expected monopoly profits motivate investment in nonrival knowledge. [Liu and Wan \(2026\)](#) study the interaction between a general-purpose AI supplier and developers of specialized applications. In their baseline model, a higher markup on general-purpose AI services can strengthen the supplier’s research incentive while raising research costs for application developers. I model a single AI developer, without distinguishing general-purpose AI from specialized applications.

The distinction between temporary and permanent growth gains relates to the production bottlenecks studied by [Aghion et al. \(2019\)](#) and [Jones and Tonetti \(2026\)](#): tasks that remain dependent on human labor can constrain growth despite automation elsewhere. In my model, RSI need not eliminate the labor bottleneck. Conversely, when AI and labor are substitutes and AI is already sufficiently efficient, permanently faster growth can persist without RSI, since capital and compute can continue expanding together. This connects to [Restrepo \(2025\)](#), where output can keep growing with the exogenous expansion of compute once bottleneck tasks have been automated.

[Acemoglu and Restrepo \(2018\)](#) show that automation can raise the long-run wage level while reducing labor’s income share. [Ray and Mookherjee \(2022\)](#) show that capital accumulation can sustain rising wages even as labor’s share vanishes. In [Hémous and Olsen \(2022\)](#), low-skill wages grow more slowly than output, while high-skill wages keep pace with it. In my AI-dominated regime, wages grow permanently faster than in the labor-bottleneck regime, but more slowly than output per worker. The substitution elasticity determines how much of the additional output-per-worker growth translates into faster wage growth.

[Korinek and Stiglitz \(2019\)](#) distinguish the gains earned by innovators from the redistribution caused by changes in wages and other factor prices. [Korinek and Ng \(2019\)](#) show how digital innovation can increase monopoly profits as a share of income while reducing the shares paid to labor and physical capital. I likewise distinguish the AI developer’s net profit from the remuneration of physical capital, and examine how these incomes evolve alongside wages as AI efficiency improves and capital accumulates.

[Moll et al. \(2022\)](#) show that automation can raise returns to wealth and increase inequality across households. In my AI-dominated regime, a higher return to physical capital accompanies permanently faster growth. I study the division of income between labor, physical capital, and AI-industry profits, but not inequality across households with different asset holdings.

3 Model

Time is continuous. The economy contains a representative household, competitive final-good firms, and an integrated AI developer. Labor-augmenting technology A grows exogenously. AI efficiency B improves through autonomous AI research: the only current input used to raise B is research compute. The model therefore describes an economy in which there are no longer human researchers developing AI and AI has achieved recursive self-improvement (RSI). As in [Trammell and Korinek](#)

(2023), I distinguish automation of research from automation of production: human labor remains an input in final-good production.

3.1 Population and labor-augmenting technology

Population grows at the exogenous rate $n \geq 0$:

$$\dot{N} = nN, \quad n \geq 0. \quad (1)$$

Each person supplies one unit of homogeneous labor inelastically. The aggregate labor endowment is therefore N . Because autonomous research uses no human labor, all labor is employed in final production and equilibrium will impose $L = N$.

The variable A measures the level of labor-augmenting technology. One unit of labor supplies A efficiency units, so effective labor is AL . Labor-augmenting technology evolves according to

$$\dot{A} = \gamma A, \quad \gamma > 0. \quad (2)$$

3.2 Household

The representative household owns physical capital and the AI developer. Let Π denote the AI developer's net profit, which accrues to the household. The household takes the paths of the net interest rate r , the wage w in final-good units and Π as given. For a predetermined capital stock $K_0 > 0$, it chooses aggregate consumption C to solve

$$\max_{\{C_t\}_{t \geq 0}} \int_0^\infty e^{-\rho t} N_t \log(C_t/N_t) dt, \quad \rho > n, \quad (3)$$

subject to

$$\dot{K} = rK + wN + \Pi - C. \quad (4)$$

The variable r is the net real interest rate: depreciation has already been deducted from the rental return to capital. The household's Euler condition is

$$\frac{\dot{C}}{C} = n + r - \rho. \quad (5)$$

Its transversality condition is

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0. \quad (6)$$

3.3 Final-good production

Competitive firms produce the final good from physical capital K , effective labor AL , and effective AI production services X . Let $0 < \alpha < 1$ denote the output elasticity with respect to capital, $\delta \geq 0$ the depreciation rate, and $\sigma > 0$ the elasticity of substitution between effective labor and effective

AI production services. The parameters ω_L and ω_X denote the CES weights on effective labor and effective AI production services, respectively. Production is

$$Y = K^\alpha Z^{1-\alpha}, \quad (7)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}. \quad (8)$$

To illustrate this production technology, consider a researcher writing an academic paper. The researcher contributes time and expertise to formulate questions, evaluate arguments, and interpret results. AI assists with derivations, programming, and drafting, while physical capital, such as a computer, runs numerical calculations and produces figures. AI can replace some of the work the researcher would otherwise do, but it can also make the researcher's expertise more productive by helping test conjectures or communicate arguments more clearly. The researcher may revise an argument drafted by AI or ask AI to check an argument written by the researcher. A numerical result may similarly combine a question formulated by the researcher, code developed with AI assistance, and calculations executed on the computer.

Jobs comprise multiple tasks, and AI may substitute for human input in some tasks while complementing it in others (Autor, 2013; Acemoglu and Restrepo, 2019). An early model of mechanization is Zeira (1998), in which producers can use more capital to reduce the labor required to produce individual intermediate goods. I model labor and AI services as aggregate inputs, not as alternative ways of performing a single task. Jones and Tonetti (2026) show how a production process with many tasks can admit a CES-like aggregate representation. Erdil et al. (2025) distinguish automating additional tasks (the extensive margin) from increasing the services supplied in tasks that AI can already perform (the intensive margin). I focus on expansion of aggregate AI services without explicitly tracking the set of automatable tasks; the fixed CES weights should therefore not be read as fractions of tasks automated.

The complementarity case connects to the “weak links” mechanism in Aghion et al. (2019, Section 9.2.2). In their task model, automation allows capital to replace labor within individual tasks, while different tasks remain complements. Tasks still performed by humans can therefore constrain production as automated tasks expand. With $\sigma < 1$, Equation (8) captures a related constraint: more AI services cannot eliminate the need for effective labor. I apply this logic to aggregate inputs with fixed weights, rather than modeling an expanding set of automated tasks; the two substitution elasticities are not interchangeable.

Let p_X denote the price of one unit of effective AI production services. Taking the input prices $r + \delta$, w , and p_X as given, a final-good firm solves

$$\max_{K, L, X \geq 0} \left\{ K^\alpha Z(AL, X)^{1-\alpha} - (r + \delta)K - wL - p_X X \right\}. \quad (9)$$

Define s_X as the elasticity of Z with respect to X :

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}. \quad (10)$$

The first-order conditions are

$$r = \alpha \frac{Y}{K} - \delta, \quad (11)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (12)$$

$$p_X = (1 - \alpha)s_X \frac{Y}{X}. \quad (13)$$

Constant returns and these conditions imply

$$Y = (r + \delta)K + wL + p_X X. \quad (14)$$

Solving the final-good firm's AI condition for its price yields the inverse demand schedule $p_X(X)$. The inverse absolute elasticity is

$$e_X \equiv -\frac{\partial \ln p_X(X)}{\partial \ln X} = \frac{1 - s_X}{\sigma} + \alpha s_X. \quad (15)$$

This demand schedule is an input to the developer's problem.

3.4 The AI developer

3.4.1 Effective AI production and research services

I use *compute* to denote the computational resources used to execute the numerical operations required by AI systems (Sastry et al., 2024). I assume that one unit of compute requires one unit of the final good, which could otherwise be consumed or invested in physical capital. Compute is therefore a rival, costly input, not a measure of AI performance.

I distinguish compute used to deploy AI from compute used to improve it. Inference compute U runs existing AI systems and produces outputs used in current final-good production. Research compute M is allocated to the development of better AI. It includes training and fine-tuning, but also computational experimentation, model evaluation, and other compute-intensive components of AI research. I therefore use the term *research compute* rather than the narrower term *training compute*. Korinek and McKelvey (2026) make a related distinction between compute used for inference and compute used for R&D and training. Compute can be measured in floating-point operations; because U and M are flows, their physical counterparts are operations per unit of time.

The distinction between U and M concerns the economic use of compute, rather than a strict separation between inference and training: running AI systems that conduct research also requires inference compute (Davidson et al., 2026).

Raw compute does not by itself determine the quantity of AI services produced. The variable B measures AI efficiency: AI services obtained per unit of compute. Thus U and M are raw computational inputs, while BU and BM are the corresponding flows of AI production and AI research services. This interpretation parallels measures of algorithmic efficiency based on the decline in compute required to achieve a given level of performance (Hernandez and Brown, 2020; Erdil and Besiroglu, 2022). A rise in B therefore represents an improvement in the algorithms that use compute, rather than an expansion of physical computing capacity. I assume that the same B augments compute in both activities.

This distinction also underlies *effective compute*, which adjusts physical compute for algorithmic improvements (Ho et al., 2024; Jones, 2026). In BM , M is rival, whereas the algorithmic knowledge represented by B is nonrival: using it in one computation does not diminish its usefulness in another. I apply this distinction to AI research more broadly than the model-training activities measured in those sources. Erdil et al. (2025) similarly distinguish the nonrival model weights created by training from the recurring compute costs of inference.

Inference compute produces AI production services according to

$$X = BU. \tag{16}$$

Equation (16) maps inference compute into the flow $X = BU$ of AI production services. Research compute analogously produces BM , the flow of AI research services.

Figure 1 summarizes the production block.

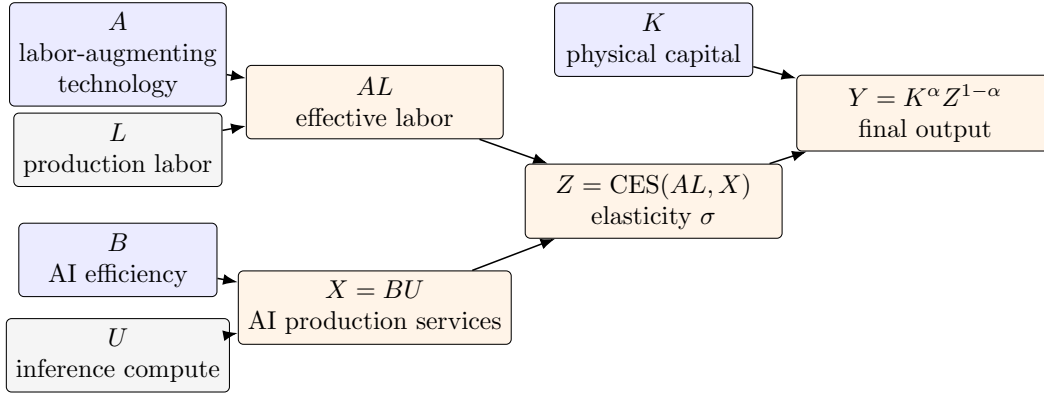


Figure 1: Production of final output and AI production services. AI efficiency B augments inference compute U to produce $X = BU$, the flow of AI production services. These services combine with effective labor AL to produce Z , which combines with capital K to produce Y .

3.4.2 RSI technology

Given initial AI efficiency $B_0 > 0$ and an upper bound $\bar{B} \in (B_0, \infty]$, AI efficiency evolves according to the RSI technology

$$\dot{B} = \chi(BM)^\eta \psi(B), \quad \psi(B) = 1 - \frac{B}{\bar{B}}, \quad \chi > 0, \quad 0 < \eta < 1. \quad (17)$$

Figure 2 summarizes the RSI mechanism.

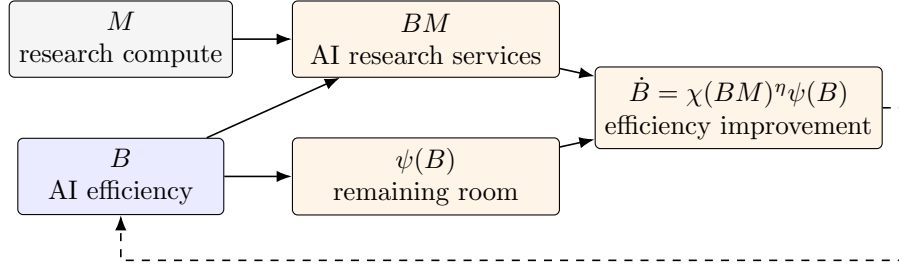


Figure 2: Recursive self-improvement in AI efficiency. AI efficiency B augments research compute M to produce BM , the flow of AI research services. These services and the remaining room for improvement $\psi(B)$ determine $\dot{B}$, which increases B .

The parameter χ shifts research productivity. The parameter η governs returns to current research scale. The same AI-efficiency index B augments both inference and research compute. This common-productivity assumption is parsimonious but substantive. Equation (17) captures RSI: a higher B produces more AI research services BM from a given amount of research compute M , and these services help improve B further. This is the AI-research feedback studied by Davidson et al. (2026). The term $\psi(B)$ reduces the remaining room for improvement as AI efficiency approaches $\bar{B}$. A related approach appears in Erdil et al. (2025), who also reduce research productivity as hardware and software efficiency approach their respective upper bounds. The uncapped specification is recovered at $\bar{B} = \infty$.

Evidence that ideas become harder to find motivates allowing research productivity to decline as technology advances. Bloom et al. (2020) document that research effort has risen substantially while research productivity has fallen across several industries, products, and firms. This evidence motivates diminishing returns, but it does not establish a literal finite technological ceiling. The term $1 - B/\bar{B}$ is a deliberately stronger reduced-form specification. With this upper bound, I establish sufficient conditions for equilibrium existence and compute equilibrium trajectories. I then examine how these equilibria change as the bound rises and whether they have a meaningful limit as $\bar{B} \rightarrow \infty$. Without the upper bound, recursive improvement may lead to the absence of an equilibrium, as shown further below.

3.4.3 Supply of effective AI production services

An integrated AI developer has exclusive ownership of B . It conducts research and sells effective AI production services. The developer sells X and purchases the $U = X/B$ units of inference compute needed to supply those services. Because X does not enter the RSI technology in equation (17) directly, its within-date choice can be solved conditional on B . Define optimized operating profit before research expenditure as

$$\pi(B) = \max_{X \geq 0} \left\{ p_X(X)X - \frac{X}{B} \right\}. \quad (18)$$

The notation $\pi(B)$ leaves its dependence on current K, A , and L implicit. The first-order condition equates marginal revenue to the compute cost of one additional service unit:

$$p_X(1 - e_X) = \frac{1}{B}. \quad (19)$$

Lemma 1 in Appendix A shows that problem (18) has a unique, strictly positive global maximizer characterized by condition (19). Thus, the developer charges the unit cost $1/B$ multiplied by the monopoly markup $1/(1 - e_X)$. The envelope theorem gives

$$\pi_B = \frac{X}{B^2} = \frac{U}{B}. \quad (20)$$

At a fixed X , a marginal increase in AI efficiency saves U/B units of the consumption good.

AI efficiency is nonrival in use. Once B has been developed, using it to augment one unit of inference or research compute does not reduce its ability to augment other units. This does not make effective AI services costless: BU and BM require the rival and costly compute inputs U and M . Nonrivalry also does not rule out excludability. An owner can restrict access to model weights, software, or AI services.

To see the resulting appropriability problem, suppose that competitive service providers had free access to B . Free entry would imply no operating rent. A developer that paid for M would improve a common AI-efficiency index B that competing providers could use without compensating it. The developer would bear the research cost but could not appropriate the resulting benefits. Exclusive ownership allows the developer to capture part of the value created by a higher B and makes the return to research appropriable, as in the standard endogenous-growth treatment of nonrival ideas (Romer, 1990).

That being said, monopoly is an institutional assumption, not a technological implication of nonrivalry. Licensing, oligopoly, public funding, or open-source provision with research subsidies could also support investment in B . Each alternative would require an additional institution and would change prices, profits, and the distribution of rents. Recent work studies these alternatives in more detail (Korinek and Vipra, 2025; Gans, 2024; Demirer et al., 2025). The single-developer assumption excludes the conflict between general-purpose AI suppliers and application developers

studied by [Liu and Wan \(2026\)](#).

3.4.4 The dynamic problem

The AI developer uses M for research and owns B and any resulting improvements in AI efficiency. After the within-date service choice, the developer's net profit is $\Pi_t = \pi(B_t) - M_t$: operating profit less research expenditure. Taking the paths of r, K, A , and L as given and substituting the static solution to problem (18), the developer chooses the path of research compute M to increase AI efficiency B :

$$\max_{\{M_t\}_{t \geq 0}} \int_0^\infty \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt, \quad (21)$$

subject to the RSI technology given by (17) and taking B_0 as given.

Let q be the current-value shadow price of AI efficiency, measured in units of the current final good per unit of B . An interior choice of research compute M satisfies

$$q\chi\eta B(BM)^{\eta-1}\psi(B) = 1, \quad (22)$$

Equation (22) equates the marginal benefit of research compute to its unit cost. Holding current B fixed, an additional unit of M raises the rate of improvement in AI efficiency by $\chi\eta B(BM)^{\eta-1}\psi(B)$. Multiplication by q values the resulting efficiency gain, accounting for future profit gains and further improvements in AI efficiency.

The current-value costate equation is

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (23)$$

Rearranging gives the return equation

$$rq = \frac{U}{B} + q\eta \frac{\dot{B}}{B} + q\dot{B} \frac{\psi'(B)}{\psi(B)} + \dot{q}. \quad (24)$$

The first term on the right-hand side of Equation (24) is the inference-compute saving from greater AI efficiency. The second is the research-productivity gain from RSI. With an upper bound, the third term is negative because a higher B leaves less room for further improvement. The final term is the capital gain on AI efficiency. Together these returns must equal the market return rq .

The developer's transversality condition is

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (25)$$

It rules out a path that leaves positive discounted value in the AI-efficiency state.

The preceding conditions are necessary for an interior optimum but do not establish global optimality. Lemma 2 in Appendix A gives sufficient conditions for a research path to maximize the

present value of the developer's net profit in problem (21).

3.5 Equilibrium

I can now combine the decisions of the three agents and the market clearing conditions to define an equilibrium. Capital-market clearing identifies household capital with the capital rented by final-good firms. Labor-market clearing gives $L = N$. AI-service market clearing identifies the developer's sales with the final sector's purchases. Combining the household budget, final-good zero profit, developer profit, and market clearing yields the aggregate resource constraint

$$C + \dot{K} + \delta K + U + M = Y. \quad (26)$$

Definition 1 (Equilibrium). *For exogenous initial values $A_0, N_0, K_0, B_0 > 0$, an equilibrium is an infinite-horizon trajectory*

$$\{K_t, B_t, C_t, q_t, L_t, U_t, M_t, X_t, Z_t, Y_t, r_t, w_t, p_{X,t}, s_{X,t}, e_{X,t}, \Pi_t\}_{t \geq 0} \quad (27)$$

such that the developer solves problem (21) and the following conditions hold:

1. The intratemporal allocation and prices satisfy

$$L = N, \quad (28a)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}, \quad (28b)$$

$$Y = K^\alpha Z^{1-\alpha}, \quad (28c)$$

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}, \quad (28d)$$

$$e_X = \frac{1 - s_X}{\sigma} + \alpha s_X, \quad (28e)$$

$$r = \alpha \frac{Y}{K} - \delta, \quad (28f)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (28g)$$

$$p_X = (1 - \alpha) s_X \frac{Y}{X}, \quad (28h)$$

$$\frac{1}{B} = p_X (1 - e_X), \quad (28i)$$

$$X = BU, \quad (28j)$$

$$1 = q_X \eta B (BM)^{\eta-1} \psi(B), \quad (28k)$$

$$\Pi = p_X X - U - M. \quad (28l)$$

2. The four dynamic equations are

$$\dot{K} = Y - C - U - M - \delta K, \quad (29a)$$

$$\dot{B} = \chi(BM)^\eta \psi(B), \quad (29b)$$

$$\frac{\dot{C}}{C} = n + r - \rho, \quad (29c)$$

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (29d)$$

3. The two transversality conditions are

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0, \quad (30a)$$

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (30b)$$

Remark 1 (The economy without AI). With $\omega_X = 0$ and $\omega_L = 1$, the model reduces to the Ramsey–Cass–Koopmans economy with logarithmic utility and exogenous labor-augmenting technological progress.

The system contains no separate household budget or final-good zero-profit equation because these are redundant after market clearing. The consolidated income decomposition follows from the equations in the definition:

$$1 = \frac{(r + \delta)K}{Y} + \frac{wL}{Y} + \frac{\Pi}{Y} + \frac{U}{Y} + \frac{M}{Y}. \quad (31)$$

It records how final output is divided among gross capital income, labor income, developer profit, inference compute, and research compute.

4 Equilibrium with an upper bound on AI efficiency

In this section, I characterize the long-run equilibrium behavior for various parameter configurations. The key finding is that long-run output-per-worker growth can exceed the rate of labor-augmenting technological change when the elasticity of substitution between AI production services and labor exceeds one and the upper bound on AI efficiency is sufficiently high.

To identify this threshold, I calculate the net return on capital implied by the production and pricing conditions as AI efficiency approaches its upper bound and labor's income share tends to zero. I compare this return with $\rho + \gamma$, the interest rate that sustains consumption-per-person growth at the exogenous rate γ . When the former exceeds the latter and AI and labor are substitutes, the model supports equilibrium paths along which capital and AI production services grow faster than effective labor, even though AI efficiency converges to a finite limit.

4.1 The return to capital and the efficiency threshold

An upper bound restricts AI efficiency, but not the compute used to supply AI production services. To identify the return when labor's income share vanishes, fix $B > 0$ and consider $s_X \rightarrow 1$. Since $wL/Y = (1 - \alpha)(1 - s_X)$, this is the zero-labor-share limit.

Combine the final-good firm's demand for AI services in equation (13) with the developer's pricing condition in equation (19). The result is:

$$\frac{X}{Y} = B(1 - \alpha)s_X(1 - e_X) \longrightarrow B(1 - \alpha)^2. \quad (32)$$

The limit uses $e_X \rightarrow \alpha$ from Equation (15). The CES aggregator in equation (8) and the definition of s_X in equation (10) imply

$$\frac{Z}{X} = \left(\frac{\omega_X}{s_X}\right)^{\sigma/(\sigma-1)} \longrightarrow \omega_X^{\sigma/(\sigma-1)}. \quad (33)$$

Using $Z/Y = (Z/X)(X/Y)$ and the limits in equations (32) and (33), I obtain

$$\frac{Y}{K} = \left(\frac{Z}{Y}\right)^{(1-\alpha)/\alpha} \longrightarrow \left[(1 - \alpha)^2 B \omega_X^{\sigma/(\sigma-1)}\right]^{(1-\alpha)/\alpha}. \quad (34)$$

Let $\mathcal{R}(B)$ be the net interest rate when $s_x \rightarrow 1$ for a given B . Substituting the limiting output-capital ratio in equation (34) into $r = \alpha Y/K - \delta$ gives

$$\mathcal{R}(B) \equiv \alpha \left[(1 - \alpha)^2 B \omega_X^{\sigma/(\sigma-1)}\right]^{(1-\alpha)/\alpha} - \delta. \quad (35)$$

This is a return at a particular limit of the production and pricing conditions, not necessarily an equilibrium interest rate. For $\sigma > 1$, $s_X \rightarrow 1$ means that AI production services become arbitrarily large relative to effective labor. For $\sigma < 1$, $s_X \rightarrow 1$ means that effective labor services become arbitrarily large relative to AI production services. Whether the equilibrium interest rate converges to $\mathcal{R}(\bar{B})$ depends on both σ and $\bar{B}$, as shown below.

By the Euler equation (5), consumption per person grows at $r - \rho$. A long-run equilibrium in which consumption per person grows at γ therefore requires $r = \rho + \gamma$. Define $\mathcal{B}(\sigma)$ as the AI-efficiency level at which the zero-labor-share return equals $\rho + \gamma$, the return required for consumption per person to grow at γ :

$$\begin{aligned} \mathcal{R}(\mathcal{B}(\sigma)) &= \rho + \gamma, \\ \mathcal{B}(\sigma) &= \frac{[(\rho + \gamma + \delta)/\alpha]^{\alpha/(1-\alpha)}}{(1 - \alpha)^2 \omega_X^{\sigma/(\sigma-1)}}. \end{aligned} \quad (36)$$

The inequalities between $\bar{B}$ and $\mathcal{B}(\sigma)$ —whether AI can attain sufficient efficiency— and between σ and 1—whether labor and AI are substitutes or complements— distinguish the long-run equilibrium regimes characterized below. As [Duffy and Papageorgiou \(2000, Section 2.2\)](#) explain for CES models of capital and labor, an elasticity above one permits sustained growth only if the limiting return on

capital is sufficiently high.

4.2 Equilibrium existence and long-run regimes

Define $g_{Y/N}$ as the growth rate of output per worker and g_w as the wage growth rate. The next proposition characterizes the long-run behavior of equilibrium trajectories.

Proposition 1 (Equilibrium existence and long-run characterization). *Fix $0 < \bar{B} < \infty$. For each parameter configuration covered by a case below, there exists a nonempty open set of initial conditions (A_0, N_0, K_0, B_0) such that every initial state in that set has an equilibrium trajectory with the following properties:*

(i) **Complementarity.** *If $0 < \sigma < 1$ and $\bar{B} > \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} s_X \in (0, 1). \quad (37)$$

(ii) **Unit elasticity.** *If $\sigma = 1$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad s_X = \omega_X \quad (38)$$

(iii) **Substitution with a labor bottleneck.** *If $\sigma > 1$ and $\bar{B} < \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} s_X \in (0, 1). \quad (39)$$

(iv) **AI-dominated growth.** *If $\sigma > 1$ and $\bar{B} > \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \mathcal{R}(\bar{B}) - \rho > \gamma, \quad (40)$$

$$g_w \rightarrow \gamma + \frac{\mathcal{R}(\bar{B}) - \rho - \gamma}{\sigma} > \gamma, \quad (41)$$

$$r \rightarrow \mathcal{R}(\bar{B}), \quad s_X \rightarrow 1.$$

Proof. See Appendix A, [Proof of Proposition 1](#). □

I treat $0 < \sigma < 1$ with $\bar{B} < \mathcal{B}(\sigma)$ separately because the result established for this region is conditional, not an existence result. Proposition 7 in Appendix A shows that any equilibrium in this region has an AI production bottleneck: long-run average output-per-worker growth is below γ , and labor's income share converges to zero.

The labor-bottleneck cases of Proposition 1 show that autonomous AI research need not eliminate the production bottleneck, a constraint also emphasized by [Aghion et al. \(2019\)](#). Moreover, $\sigma > 1$ is not sufficient to escape it: if $\mathcal{R}(\bar{B}) < \rho + \gamma$, the return at a vanishing labor share is too low to sustain growth faster than effective labor.

The AI-dominated regime has an asymptotically linear production structure, as in the capital-accumulation mechanisms of [Rebelo \(1991\)](#) and [Jones and Manuelli \(1990\)](#). Equation (34) gives a positive limiting output–capital ratio even as B approaches the upper bound. Capital and inference compute can expand together, sustaining growth above γ even when B is arbitrarily close to $\bar{B}$ and further AI-efficiency gains vanish, as in the compute-expansion mechanism of [Restrepo \(2025\)](#). [Erdil et al. \(2025\)](#) also discuss how expanding digital-worker services alongside physical capital can generate AK -type dynamics. Here, the corresponding growth regime arises under fixed aggregate production weights and private research and pricing decisions. RSI determines how efficiency improves toward a level at which this mechanism supports growth faster than γ .

This joint expansion also explains the higher long-run interest rate. In the labor-bottleneck regime, the interest rate converges to $\rho + \gamma$. In the AI-dominated regime, however, AI services expand alongside capital, so production need not be constrained by the growth of effective labor. The output–capital ratio converges to a level at which $r = \alpha Y/K - \delta$ remains above $\rho + \gamma$: accumulating more capital does not restore the labor-bottleneck return. [Caselli and Manning \(2019\)](#) also discuss the possibility that technological progress raises growth and the interest rate together, but their comparison of steady states does not determine the resulting wage dynamics.

The proof of Proposition 1 first normalizes growing quantities and shrinking efficiency gaps. For each parameter configuration covered by the proposition, I conjecture stationary values of these normalized variables. I then construct trajectories from every sufficiently nearby initial state, deriving equilibrium trajectories and checking they converge to the stationary conjectures. Local uniqueness results are given in the proof. These analytical results guide the algorithm’s boundary conditions (Appendix B).

Remark 2 (Wages and AI revenue in the AI-dominated regime). In the AI-dominated equilibria of Proposition 1(iv), output per worker grows asymptotically faster than wages, even though both grow faster than labor-augmenting technological progress:

$$\lim_{t \rightarrow \infty} g_{Y/N} > \lim_{t \rightarrow \infty} g_w > \gamma.$$

Labor’s income share therefore converges to zero, while the AI industry’s revenue as a share of output converges to $1 - \alpha$.

Rising wages alongside a declining labor share also appear in [Acemoglu and Restrepo \(2018\)](#), [Autor and Kausik \(2026\)](#), and [Ray and Mookherjee \(2022\)](#). In my AI-dominated regime, wages grow permanently faster than in the labor-bottleneck regime, even as AI efficiency approaches its upper bound. Equations (40) and (41) quantify this increase: relative to the labor-bottleneck regime, the increase in wage growth equals the increase in output-per-worker growth divided by the elasticity of substitution.

Figure 3 shows where the long-run regimes characterized in Proposition 1 lie relative to $\mathcal{B}(\sigma)$. For $0 < \sigma < 1$, the equilibria above the threshold have output-per-worker growth converging to γ and a positive limiting labor income share. For $\sigma > 1$, these same long-run outcomes occur below

the threshold. Above it, the regime is AI-dominated: output per worker grows asymptotically faster than γ and labor's income share converges to zero.

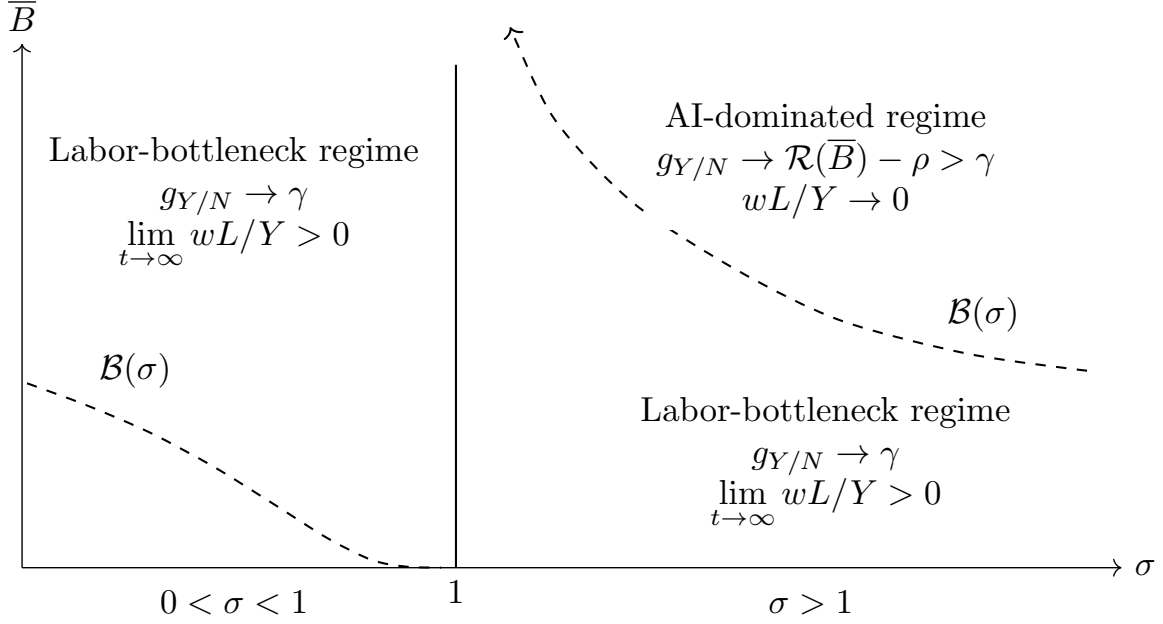


Figure 3: Growth regimes with an upper bound on AI efficiency.

Table 1 summarizes the four cases of Proposition 1. For $\sigma > 1$, an upper bound below $\mathcal{B}(\sigma)$ makes the return on capital at a vanishing labor share too low for capital and AI services to outgrow effective labor. Above the threshold, that return supports their faster growth along the AI-dominated equilibria. Across these equilibrium regimes, long-run wage growth exceeds γ if and only if labor's income share converges to zero.

Table 1: Long-run equilibrium characterization

Elasticity	AI efficiency bound	$\lim g_{Y/N}$	$\lim g_w$	$\lim wL/Y$	$\lim r$
$0 < \sigma < 1$	$\bar{B} > \mathcal{B}(\sigma)$	γ	γ	Positive	$\rho + \gamma$
$\sigma = 1$	Any finite $\bar{B}$	γ	γ	$(1 - \alpha)\omega_L$	$\rho + \gamma$
$\sigma > 1$	$\bar{B} < \mathcal{B}(\sigma)$	γ	γ	Positive	$\rho + \gamma$
$\sigma > 1$	$\bar{B} > \mathcal{B}(\sigma)$	$\mathcal{R}(\bar{B}) - \rho$	$\gamma + \frac{\mathcal{R}(\bar{B}) - \rho - \gamma}{\sigma}$	Zero	$\mathcal{R}(\bar{B})$

4.3 Monopoly pricing and the growth threshold

The threshold in equation (36) reflects both production technology and monopoly pricing. To isolate the effect of pricing, I hold AI efficiency fixed and compare the monopolist's choice with AI services supplied at marginal cost, $p_X = 1/B$.

Proposition 2 (Monopoly pricing and the AI-efficiency threshold). *Suppose $\sigma > 1$. Under marginal-cost pricing, let $\mathcal{R}^{\text{mc}}(B)$ denote the net return on capital implied by production and pricing as $s_X \rightarrow 1$. For every $B > 0$,*

$$\mathcal{R}^{\text{mc}}(B) = \alpha \left[(1 - \alpha) B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha} - \delta > \mathcal{R}(B). \quad (42)$$

The efficiency threshold $\mathcal{B}^{\text{mc}}(\sigma)$ defined by $\mathcal{R}^{\text{mc}}(\mathcal{B}^{\text{mc}}(\sigma)) = \rho + \gamma$ satisfies

$$\mathcal{B}^{\text{mc}}(\sigma) = (1 - \alpha)\mathcal{B}(\sigma) < \mathcal{B}(\sigma). \quad (43)$$

Proof. See Appendix A, [Proof of Proposition 2](#). □

To understand why monopoly pricing raises the threshold, I compare the same AI technology with the same capital and effective labor under the two pricing rules. At marginal cost, AI services are cheaper, so final-good firms use more of them and produce more with the same capital. This raises the output–capital ratio and therefore the return on capital. For the threshold comparison, I evaluate both returns as labor’s income share tends to zero. In this limit, monopoly pricing requires a higher AI efficiency for the return on capital to reach the benchmark $\rho + \gamma$. With $\alpha = 0.33$, marginal-cost pricing reaches this benchmark at only 67.0% of the AI efficiency required under monopoly pricing.

The comparison also identifies a gain from wider use of an existing AI technology. Holding K , AL , and B fixed, marginal-cost pricing maximizes $Y - X/B$: output net of the inference compute needed to supply AI services. It therefore leaves more resources available for consumption, capital accumulation, or research. Yet marginal-cost pricing also eliminates the operating profits that reward research in the model. The incentive to remove a markup after a technology has been developed must therefore be considered alongside its effect on the incentive to develop that technology.

This raises a classic time-inconsistency problem ([Kydlund and Prescott, 1977](#)). A government may promise monopoly protection to encourage research, but later wish to remove the markup once the technology has been developed. Anticipating this policy reversal can weaken the developer’s incentive to invest in research. In the present setting, continued RSI makes the trade-off relevant even after a valuable AI technology already exists: opening access to that technology changes the rewards for improving it further.

An announced rule that ends monopoly protection at a specified efficiency level is different from reneging on a promise. It nevertheless changes the developer’s problem. Without compensation, reaching that level would remove future monopoly rents, potentially giving the developer an incentive to delay reaching it. The threshold $\mathcal{B}^{\text{mc}}(\sigma)$ is not an optimal intervention threshold: it identifies a return consistent with growth at γ , not the best time to change policy. Choosing that time requires accounting for the research response and comparing consumption paths under alternative rules.

Patent buyouts offer a possible way to reward innovation without retaining the monopoly markup. [Kremer \(1998\)](#) studies government purchases of patents that are generally placed in the public domain. Applied here, the idea would be to compensate the developer for the rights to an existing AI technology and allow competing suppliers to use it. Competitive supply at the

same efficiency would require access to the algorithms and model weights. The buyout would also need a credible valuation and payment rule, a source of financing, and arrangements to reward subsequent improvements. As Kremer emphasizes, incentives for further development depend on whether developers can appropriate its benefits. These issues are particularly important for RSI, where today's technology is also an input into tomorrow's research. I leave the design of such a policy, including its effects on research and welfare, to an extension of the model.

5 Equilibrium without an upper bound on AI efficiency

I now remove the upper bound on AI efficiency and compare the three substitution regimes. The RSI technology becomes

$$\dot{B} = \chi(BM)^\eta. \quad (44)$$

All other primitives remain as in Section 3. In Definition 1, I set $\psi(B) = 1$ and $\psi'(B) = 0$ and remove the upper restriction on B . Equilibrium still requires finite allocations at every finite date, global agent optimality, and both transversality conditions.

The results differ in scope across substitution regimes. Under complementarity, long-run average output-per-worker growth cannot exceed the rate of labor-augmenting technological progress, even without an upper bound on AI efficiency. At unit elasticity, I give sufficient parameter restrictions for an analytically tractable balanced-growth equilibrium. Under substitution and sufficiently strong returns to research, the developer can make discounted net profit arbitrarily large even over a fixed finite horizon, ruling out an equilibrium with a finite-valued developer optimum.

5.1 Complementarity

When $\sigma < 1$, effective labor remains necessary even if AI becomes arbitrarily efficient. Equation (8) gives an upper bound for Z and Y which does not depend on B :

$$Z \leq \omega_L^{\sigma/(\sigma-1)} AL, \quad (45)$$

$$Y \leq \omega_L^{\sigma(1-\alpha)/(\sigma-1)} K^\alpha (AL)^{1-\alpha}. \quad (46)$$

Proposition 3 (Production bounds under complementarity). *Let $0 < \sigma < 1$ and $\bar{B} = \infty$. Along any feasible path satisfying the production equations (7) and (8), labor-market clearing $L = N$ in equation (28a), and the resource constraint (26), the following holds:*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \gamma. \quad (47)$$

Proof. See Appendix A, Proof of Proposition 3. □

Complementarity suffices to bound long-run growth despite RSI and no bound on AI efficiency.

Proposition 4 (Long-run outcomes under complementarity without an AI efficiency upper bound). *Suppose $0 < \sigma < 1$, $\bar{B} = \infty$, and an equilibrium satisfies $B \rightarrow \infty$. Then*

$$\frac{K}{Y} \rightarrow \frac{\alpha}{\rho + \gamma + \delta} > 0, \quad \frac{C}{Y} \rightarrow 1 - \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} > 0, \quad (48)$$

$$\frac{U}{Y} \rightarrow 0, \quad \frac{M}{Y} \rightarrow 0, \quad (49)$$

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad (50)$$

$$s_X \rightarrow \frac{1 - \sigma}{1 - \alpha\sigma}, \quad (51)$$

$$\frac{wL}{Y} \rightarrow \frac{\sigma(1 - \alpha)^2}{1 - \alpha\sigma} > 0, \quad \frac{p_X X}{Y} \rightarrow \frac{(1 - \alpha)(1 - \sigma)}{1 - \alpha\sigma} > 0. \quad (52)$$

Proof. See Appendix A, Proof of Proposition 4. $\square$

Inference and research compute vanish as shares of output. The AI industry nevertheless retains a positive revenue share.

5.2 Unit elasticity

At unit elasticity, production becomes

$$Y = K^\alpha (AL)^{(1-\alpha)\omega_L} X^{(1-\alpha)\omega_X}. \quad (53)$$

Consider a balanced-growth path (BGP) with positive constant K/Y , C/Y , and M/Y . The AI-demand and monopoly conditions, equations (13) and (19), imply $U/Y = (1 - \alpha)^2 \omega_X^2$. Along the BGP, $g_K = g_U = g_M = g_Y$. Substituting $X = BU$ into production, equation (53), and log-differentiating this equation and the RSI technology in equation (44) yields

$$g_Y = n + \gamma + \frac{\omega_X}{\omega_L} g_B, \quad (54)$$

$$(1 - \eta)g_B = \eta g_Y. \quad (55)$$

The first relation describes how AI improvement raises production growth. The second describes how expanding research resources sustain AI improvement. Solving these relations gives candidate growth rates. The next proposition establishes that they belong to an equilibrium.

Proposition 5 (An uncapped balanced-growth equilibrium and local convergence). *Suppose $\sigma = 1$, $\bar{B} = \infty$, and*

$$\eta + \omega_X < 1. \quad (56)$$

(i) *There is an equilibrium trajectory such that*

$$g_{Y/N} = g_w = \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X}(n + \gamma), \quad (57)$$

$$r = \rho + \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X}(n + \gamma), \quad (58)$$

$$\frac{wL}{Y} = (1 - \alpha)\omega_L > 0, \quad (59)$$

$$g_B = \frac{\eta(1 - \omega_X)}{1 - \eta - \omega_X}(n + \gamma). \quad (60)$$

(ii) *There is an open neighborhood of the initial condition (A_0, N_0, K_0, B_0) of the BGP in part (i) such that, from every initial condition in that neighborhood, there exists an equilibrium trajectory converging to the corresponding BGP in normalized variables.*

Proof. See Appendix A, Proof of Proposition 5. □

Proposition 5 shows that RSI can sustain permanently faster wage growth without a vanishing labor income share. Equations (57) and (59) imply $g_w = g_{Y/N} > \gamma$ while $wL/Y = (1 - \alpha)\omega_L$ remains constant and positive. Wages therefore keep pace with output per worker. This contrasts with the regimes with an upper bound in Proposition 1, where a permanent wage-growth premium occurs only in the AI-dominated regime and accompanies a vanishing labor share. A declining labor share is therefore not a necessary consequence of recursive self-improvement.

Growth in effective labor expands the resources available for research, which raises B and further expands output. The common denominator $1 - \eta - \omega_X$ captures this reinforcement: a higher η strengthens the research feedback, while a higher ω_X makes output growth more responsive to AI improvement. A related dependence appears in Jones (1995), where long-run growth depends on an expanding research workforce. Here, effective labor supports research indirectly through production, rather than entering the RSI technology itself.

5.3 Substitution

With $\sigma > 1$, Proposition 1 establishes AI-dominated equilibria whose limiting interest rate is $\mathcal{R}(\bar{B})$, which grows without bound as $\bar{B} \rightarrow \infty$. The same function also bounds the equilibrium interest rate without an efficiency upper bound:

$$r_t \geq \mathcal{R}(B_t). \quad (61)$$

To see this, combine equations (32), (33), and (34) before taking the limit $s_X \rightarrow 1$:

$$\frac{Y}{K} = \left[B(1 - \alpha)\omega_X^{\sigma/(\sigma-1)}(1 - e_X)s_X^{-1/(\sigma-1)} \right]^{(1-\alpha)/\alpha}. \quad (62)$$

For fixed B and $\sigma > 1$, this expression decreases with s_X and approaches its lower limit as $s_X \rightarrow 1$. Substitution into $r = \alpha Y/K - \delta$ gives inequality (61). Thus $\mathcal{R}(B)$ is the lowest return compatible

with the production and monopoly conditions at that efficiency level, attained in the zero-labor-share limit. If B grows without bound along a trajectory, inequality (61) requires $r \rightarrow \infty$ and $Y/K \rightarrow \infty$. Production also gives $Z/Y = (Y/K)^{\alpha/(1-\alpha)} \rightarrow \infty$, while the Euler equation (5) requires $g_{C/N} = r - \rho \rightarrow \infty$.

Davidson et al. (2026) show that technological and economic feedback loops can overcome diminishing returns to research and generate *hyperbolic growth*, with variables diverging at a finite date. This *finite-time singularity* is stronger than an indefinitely increasing growth rate. Their benchmark fixes saving and factor-allocation shares; I therefore cannot apply its explosion conditions directly to the optimal choices studied here.

Under Definition 1, a finite-time singularity cannot constitute an infinite-horizon equilibrium, which requires finite allocations at every finite date. The return bound in equation (61) does not determine whether divergence occurs in finite or infinite time. The following proposition instead shows that, when research returns are sufficiently strong, the developer can make discounted net profit arbitrarily large even over a fixed finite horizon. This is an unbounded optimization problem, not a trajectory that attains infinite profit in finite time.

Proposition 6 (Research expenditure and the developer’s value). *Suppose $\sigma > 1$, $\bar{B} = \infty$ and $0 < \alpha < \eta < 1$. Fix any $B_0 > 0$, any finite horizon $T > 0$, positive continuous paths of K, A, L , and an integrable interest-rate path. For the developer’s problem (21) restricted to $[0, T]$, the developer can make discounted net profit arbitrarily large.*

Proof. See Appendix A, Proof of Proposition 6. □

6 Numerical simulations

6.1 Parameters and initial conditions

In this section, I compare the transitional dynamics across four parameter configurations corresponding to the cases in Proposition 1. I study an unanticipated activation of RSI at date zero. Before date zero, the economy already uses AI production services, but AI efficiency is fixed at B_0 . For the chosen value of B_0 , each economy starts on a labor-bottleneck balanced growth path. I compute this path with the research productivity parameter χ equal to zero to obtain the corresponding initial capital stock K_0 for each parameter configuration. Setting $\chi = 0$ rules out improvements in AI efficiency, not the expansion of AI services through inference compute U . By treating activation as unanticipated, I exclude the advance adjustment of consumption and saving emphasized in the technology-news literature reviewed by Beaudry and Portier (2014).

I compare four economies with $\sigma = 0.9, 1.0, 1.1, 1.5$, holding all other parameters and the initial values A_0, N_0 , and B_0 fixed. For each economy, I choose K_0 to place it on its pre-RSI balanced growth path. In particular, for each elasticity, I choose the pre-event capital stock to satisfy

$$r_0 = \rho + \gamma, \quad \frac{K_0}{Y_0} = \frac{\alpha}{\rho + \gamma + \delta} = 3.30. \quad (63)$$

This setup guarantees that Y , X , U , w , r , and p_X behave continuously rather than jumping at date zero.

Table 2 reports the parameter values and initial stocks used for the simulations. I use standard values for α , δ , and ρ , while n and γ are guided by long-run population and productivity projections. The AI production weight $\omega_X = 0.10$ and research exponent $\eta = 0.20$ are illustrative choices, not empirical estimates. I choose them to satisfy $\omega_X + \eta < 1$, the condition for the uncapped unit-elastic balanced-growth equilibrium in Proposition 5; this is not a requirement for the simulations with an upper bound considered here. The common upper bound on AI efficiency satisfies $\mathcal{B}(1.5) < \bar{B} < \mathcal{B}(1.1)$. Thus, AI and labor are substitutes in both the $\sigma = 1.1$ and $\sigma = 1.5$ cases, but only the latter converges to the AI-dominated regime.

I study two levels of research productivity for each of the four elasticities of substitution. They both led to the same long-run growth rates in all variables, but differ in the speed at which growth accelerates once RSI becomes available. In the first one, higher-productivity scenario, χ is equal to 7.5. In the second one, the lower-productivity scenario, χ is equal to 1.5. These are also illustrative choices, but their contrast show how different parameter values can lead to significant differences in the short-run impact of AI.

Table 2: Simulation parameters and initial stocks

Symbol	Value	Interpretation	Source
α	0.33	Capital share	Standard calibration value.
δ	0.05	Depreciation rate	Standard calibration value.
ρ	0.04	Discount rate	Standard calibration value.
n	0.003	Population growth rate	Rounded constant-rate approximation to the UN 2024–2100 world projection (United Nations, 2024).
γ	0.01	Growth of labor-augmenting technology	OECD labor-productivity projections (OECD, 2025).
η	0.20	Research exponent	Illustrative.
ω_X	0.10	AI production weight	Illustrative.
ω_L	0.90	Effective-labor weight	$1 - \omega_X$.
χ	7.5 (higher); 1.5 (lower)	Research productivity after activation	Illustrative.
σ	0.90; 1.00; 1.10; 1.50.	Elasticity of substitution	Illustrative.
$\bar{B}$	1,361	AI-efficiency upper bound	$1.10 \times \mathcal{B}(1.50)$.
A_0	1	Initial labor-efficiency units	Normalization.
N_0	1	Initial population	Normalization.
B_0	13.6	Initial AI efficiency	$0.01 \times \bar{B}$.
K_0	3.43; 4.65; 5.04; 5.50.	Initial capital	Adjusted to match $K_0/Y_0 = 3.30$, the pre-RSI balanced growth path ratio.

I do not model a separate stock of computing hardware or the costs of rapidly expanding it. [Erdil et al. \(2025\)](#) incorporate adjustment costs for compute capacity and other physical capital. Introducing such frictions could change the initial reallocation of resources and the timing of the transition shown here.

Appendix [B](#) describes the algorithm used to find the equilibrium and presents the links to replication files.

6.2 Higher research productivity

I now compare the transitional dynamics for the four elasticities under relatively high research productivity, $\chi = 7.5$. Research productivity affects the transition, but not the limiting growth rates characterized in Section [4](#). All four economies inherit the pre-RSI balanced growth paths described in Section [6.1](#). Table [2](#) reports their common parameters and elasticity-specific initial capital stocks. I examine quantities first, then wages and prices, and finally income shares.

I use Figure [4](#) to present the growth of $Y/(AL)$, $X/(AL)$, and $K/(AL)$. The plots present the growth rates for output, AI production services and capital relative to effective-labor. Thus zero

means that the variable is growing at the same rate of effective labor, $n + \gamma$. The upper row shows two pre-event years and the first ten years after RSI activation. The lower row follows the same variables from year 10 to year 500. Throughout the three figures, green dashed, black solid, orange dotted, and blue dash-dotted lines denote $\sigma = 0.90, 1, 1.10, 1.50$, respectively. The vertical line marks the RSI activation, and black dotted horizontal lines mark the analytical limits for $\sigma = 1.50$.

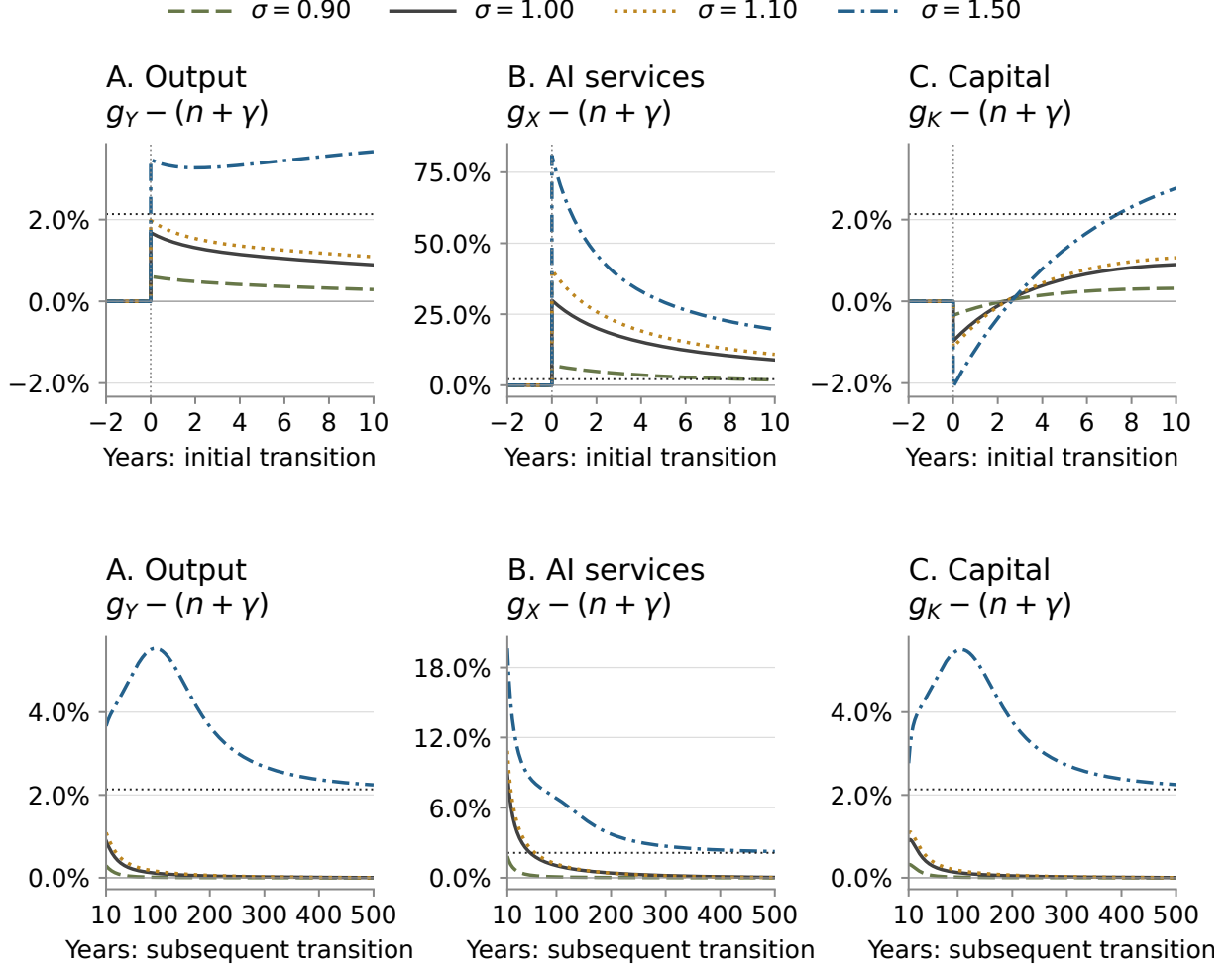


Figure 4: Higher research productivity ($\chi = 7.5$): growth of output, AI production services, and capital per unit of effective labor. Rates are instantaneous annual rates. The upper row shows two pre-event years and years 0–10; the lower row shows years 10–500. Panels A and C share a scale within each row. The vertical line marks RSI activation.

The first three panels in the top row of Figure 4 show that AI services naturally respond much more strongly than output. At $\sigma = 1.50$, $X/(AL)$ initially grows at 80.8% per year, whereas $Y/(AL)$ grows at 3.5%, which is still pretty significant. Research begins to raise B , allowing more services to be obtained from compute through $X = BU$. Final production cannot expand at the same rate because it also requires capital and labor. Initial growth of $Y/(AL)$ is smaller at the other elasticities, reaching only 0.6% at $\sigma = 0.90$. The third panel shows the capital response: $K/(AL)$

falls in every case. Households increase consumption at RSI activation while the developer starts spending on research. The resource constraint thus requires lower investment in physical capital. Capital accumulation subsequently recovers as output expands and as its marginal product increases.

The last three panels in the bottom row of Figure 4 separate the temporary growth response from the long-run growth response. At $\sigma = 0.90, 1, 1.10$, all three normalized growth rates decline toward zero. This is the labor-bottleneck outcome in Proposition 1: output per person eventually grows at $\gamma = 1.0\%$. At $\sigma = 1.50$, output and capital growth instead rise further before declining toward their long run values. As explained in Section 4.2, sufficient substitution and a sufficiently high efficiency bound allow capital and inference compute to expand together without being limited by effective-labor growth. Output, AI services and capital eventually grow 2.1 percentage points above effective labor, which always grows at 1.3% per year.

I next examine wages and prices in Figure 5. The figure shows the evolution of wage growth g_w , the net interest rate r and the level of p_X on a logarithmic scale. The top panels present the short-run dynamics and the bottom panels present the long-run dynamics. Remarkably, wage growth increases in all four economies, but it is not ordered monotonically by substitutability: the immediate increase is larger at $\sigma = 1$ than at $\sigma = 1.50$. However, it does not take long for wage growth in the highest elasticity case to rapidly surpass all other cases. Whereas in the labor-bottleneck cases wages keep growing but immediately decelerate, wages in the AI dominated scenario keep accelerating for considerable time. Meanwhile, the interest rate starts at 5.0% in every case and then rises as output initially outpaces capital. The higher return accompanies faster consumption growth, as required by the Euler condition in Equation (5). A related mechanism underlies Chow et al. (2026), who link faster expected consumption growth to higher long-term real interest rates. Meanwhile, p_X falls continuously as AI efficiency improves.

The bottom row of Figure 5 shows the long-run dynamics for the wage growth, the interest rate, and the price of AI productions services. The three labor-bottleneck economies return toward wage growth of 1.0% and an interest rate of 5.0%. At $\sigma = 1.50$, both rates rise above their long-run values before declining. Equation (41) gives limiting wage growth of 2.4%, below output-per-person growth of 3.1% but above the rate of labor-augmenting efficiency of 1.0%. Meanwhile the interest rate tend to $\mathcal{R}(\bar{B}) = 7.1\%$, above its pre-RSI level.

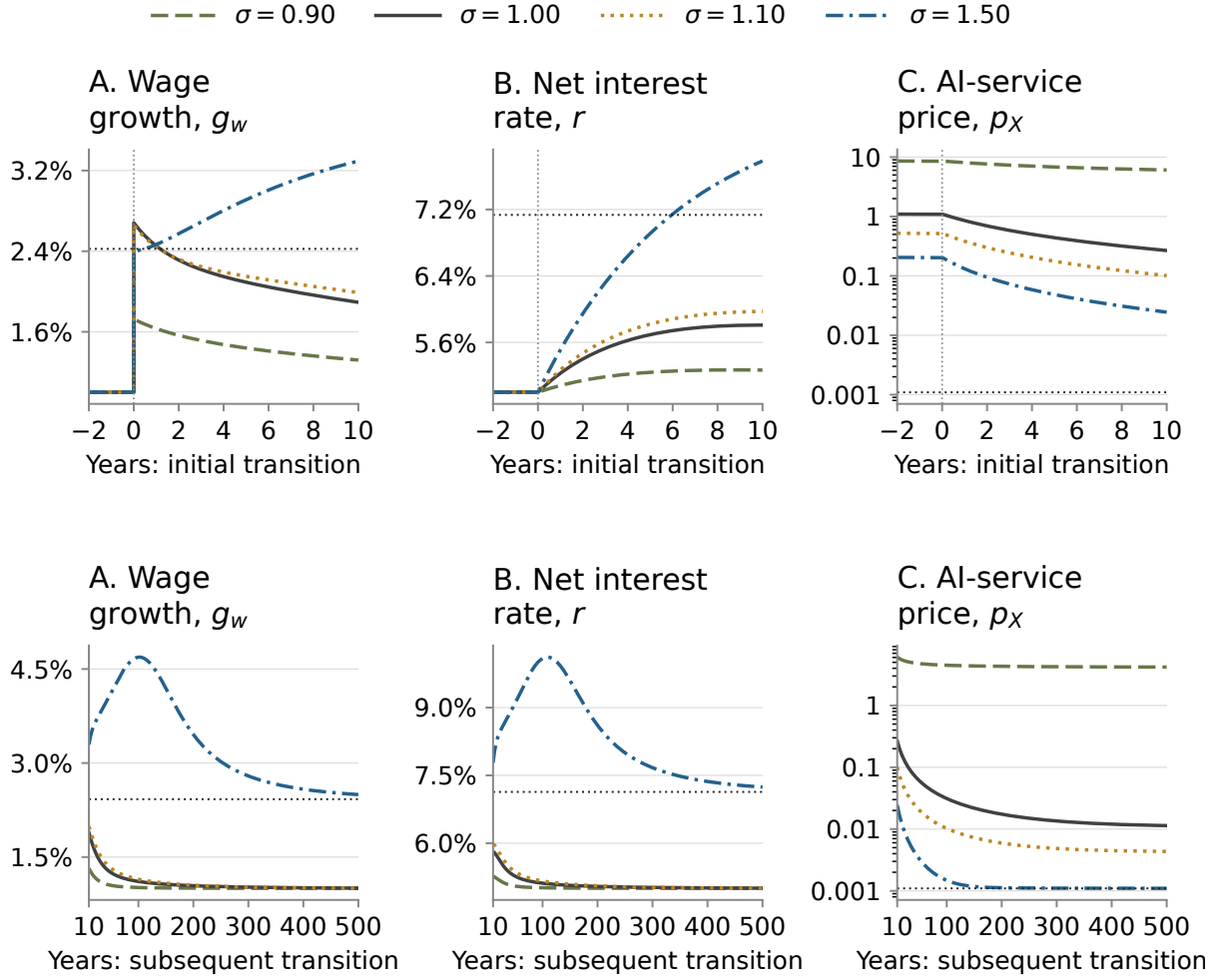


Figure 5: Higher research productivity ($\chi = 7.5$): wage growth, the net interest rate, and the AI-service price. Rates are annual; the price is in final-good units on a logarithmic scale. The vertical line marks RSI activation.

Finally, I use Figure 6 to show labor income wL/Y , developer net profit Π/Y , inference expenditure U/Y , and research expenditure M/Y , all as a share of output. The top four panels show the short run and the bottom four panels show the long run. Line styles again identify the four elasticities. The omitted gross capital income share remains $\alpha = 33.0\%$.

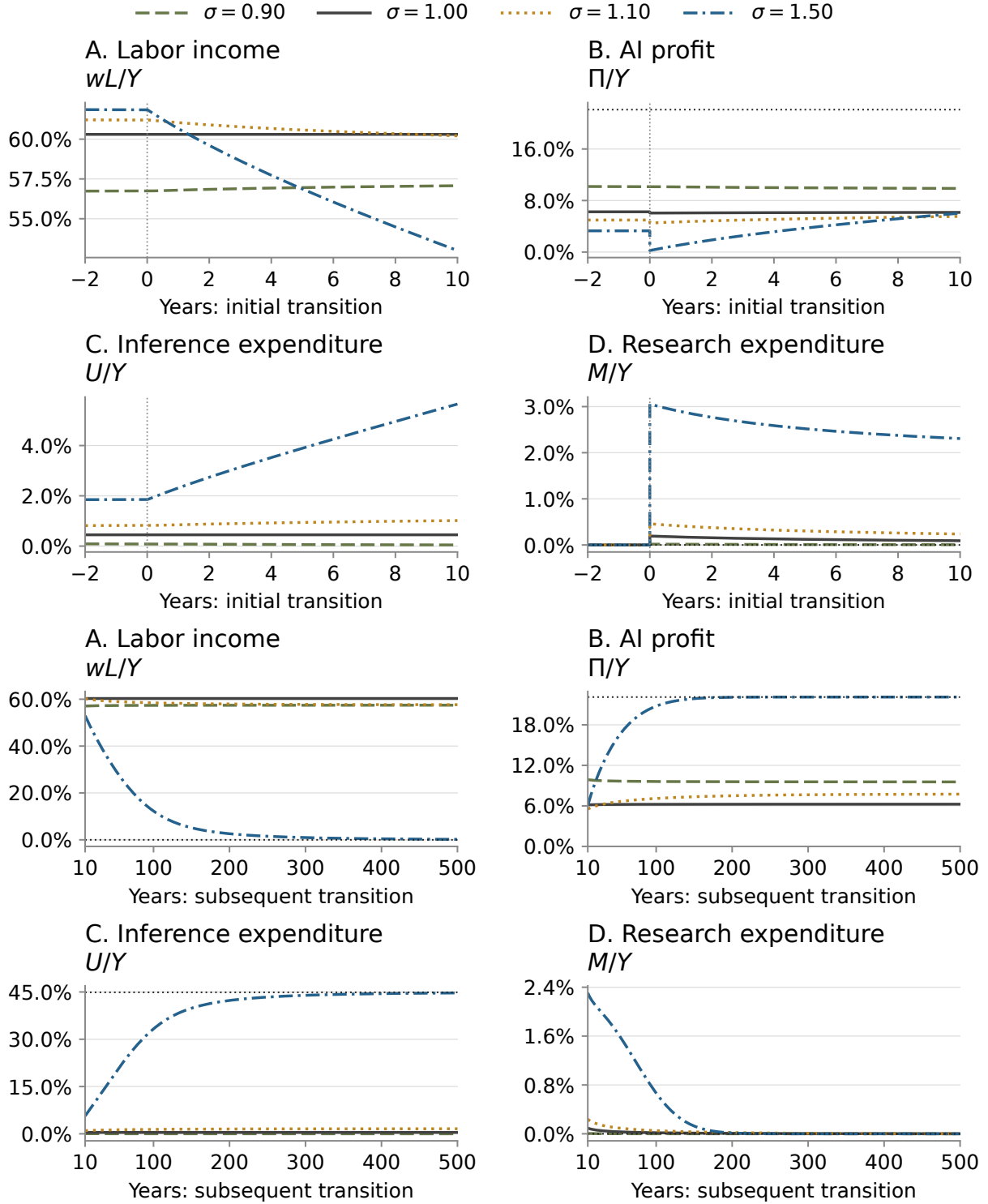


Figure 6: Higher research productivity ($\chi = 7.5$): labor income, AI net profit, inference expenditure, and research expenditure relative to output. The first two rows show the pre-event BGP and years 0–10; the last two show years 10–500. The omitted gross capital share remains $\alpha = 0.33$. Short-run panels A and C use scales fitted to the initial trajectories and omit the long-run reference lines.

In the short run, research spending as a share of output jumps in all four economies. The adjustment is strongest at $\sigma = 1.50$: immediately after activation, research absorbs 59.4% of AI revenue, compared with 35.9% for inference compute. The developer’s net profit therefore falls sharply. Profit subsequently recovers as research absorbs a smaller share of revenue and AI sales expand. Labor’s income share also falls most strongly in this scenario, while changes are much smaller in the other three economies.

Over the longer horizon, the AI-dominated economy combines permanently faster wage growth with an even faster expansion of output per worker, so labor’s income share tends to zero. Rising wages alongside a declining labor share also appear in [Autor and Kausik \(2026\)](#) and [Ray and Mookherjee \(2022\)](#). As labor’s share falls, AI revenue rises relative to output. Inference compute eventually absorbs most of this revenue: its share converges to 67.0%, while research expenditure vanishes relative to revenue.

6.3 Lower research productivity

I now set $\chi = 1.5$, one fifth of its value in Section 6.2. All other parameters and initial stocks remain unchanged, including the elasticity-specific pre-RSI capital stocks in Table 2. Consumption and research are solved again for each economy; the new exercise changes both the initial adjustment and the timing of the transition, not the long-run growth rates and steady state outcomes.

I use Figure 7 to present the growth of $Y/(AL)$, $X/(AL)$, and $K/(AL)$ in panels A–C, in percent per year. As before, zero means growth at the rate of effective labor, $n + \gamma$. The upper row shows two pre-event years and the first ten years after RSI activation. The lower row follows the same trajectories from year 10 to year 500. Throughout the three figures, line styles identify the same four elasticities as in Figure 4, and black dotted lines mark the analytical limits for $\sigma = 1.50$.

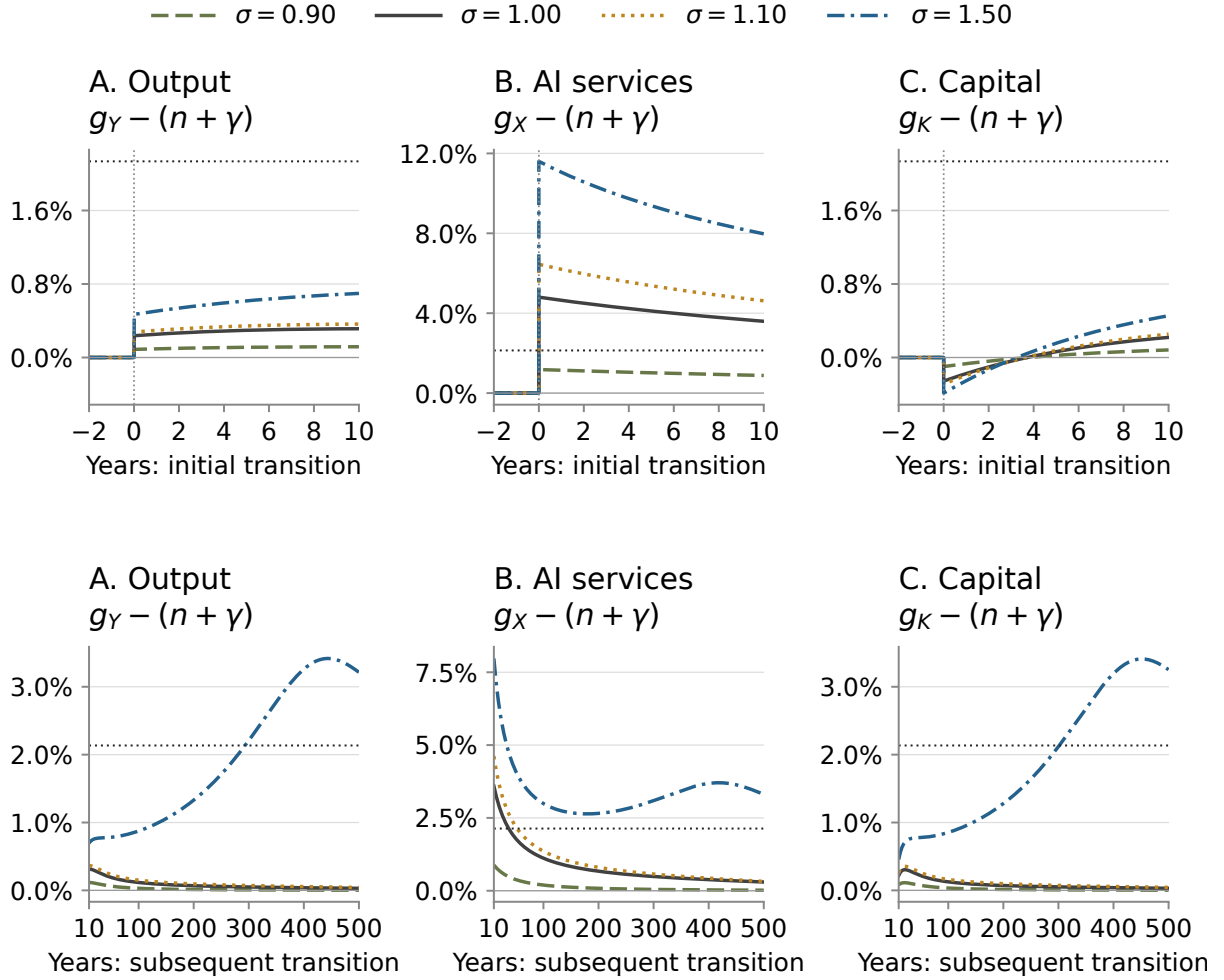


Figure 7: Lower research productivity ($\chi = 1.5$): growth of output, AI production services, and capital per unit of effective labor. Rates are instantaneous annual rates. The upper row shows two pre-event years and years 0–10; the lower row shows years 10–500. Panels A and C share a scale within each row. The vertical line marks RSI activation.

The upper row shows a much smaller initial increase in growth. Growth of $Y/(AL)$ ranges from 0.1% to 0.5% across the four economies. AI services still grow faster than output, but their initial normalized growth rate at $\sigma = 1.50$ is 11.6%, compared with 80.8% in the preceding exercise. Lower research productivity and a smaller research allocation produce slower efficiency improvements. Consumption still rises at activation, and research uses resources that could otherwise finance capital accumulation. Both adjustments are smaller, however, so $K/(AL)$ falls less. As capital accumulation recovers, output growth rises during the first decade even though AI-service growth is already slowing. The four economies therefore begin with more moderate growth and less pronounced differences.

The lower row reveals the eventual separation between regimes. At $\sigma = 1.50$, output and capital growth rise for several centuries before declining. AI-service growth first slows, then rises again

as capital and inference use expand. At year 500, $Y/(AL)$ still grows at 3.2%, above its 2.1% long-run limit, showing a significantly slower transition. A similar pattern of delayed acceleration appears in Jones (2026, pp. 11–12), drawing on Jones and Tonetti (2026). My simulations generate the transition through private RSI investment and capital accumulation, without modeling the progressive automation of additional tasks.

I next examine wages and prices in Figure 8. Panels A–C show annual wage growth g_w , the annual net interest rate r , and the level of p_X in final-good units on a logarithmic scale. The top panels show the short run and the bottom panels the long run. Initially, wage growth rises modestly in all four economies. Unlike the preceding exercise, it continues to increase during the first years even in the labor-bottleneck cases: capital accumulation helps sustain the increase in labor’s marginal product. The interest rate also rises more gradually as output outpaces capital. At $\sigma = 1.50$, it reaches 5.5% after ten years, compared with 7.8% under higher research productivity. Meanwhile, slower efficiency improvements make the decline in p_X more gradual.

In the bottom row, wage growth and the interest rate eventually turn down in all four economies. The three labor-bottleneck cases approach 1.0% and 5.0%, respectively. At $\sigma = 1.50$, the rise and subsequent decline occur much later, and adjustment remains incomplete at year 500: the interest rate is still 8.2%, above its 7.1% long run value. The limiting wage-growth rate remains 2.4%, below output-per-person growth of 3.1%. Thus the widening gap between output and wages reappears, but much later.

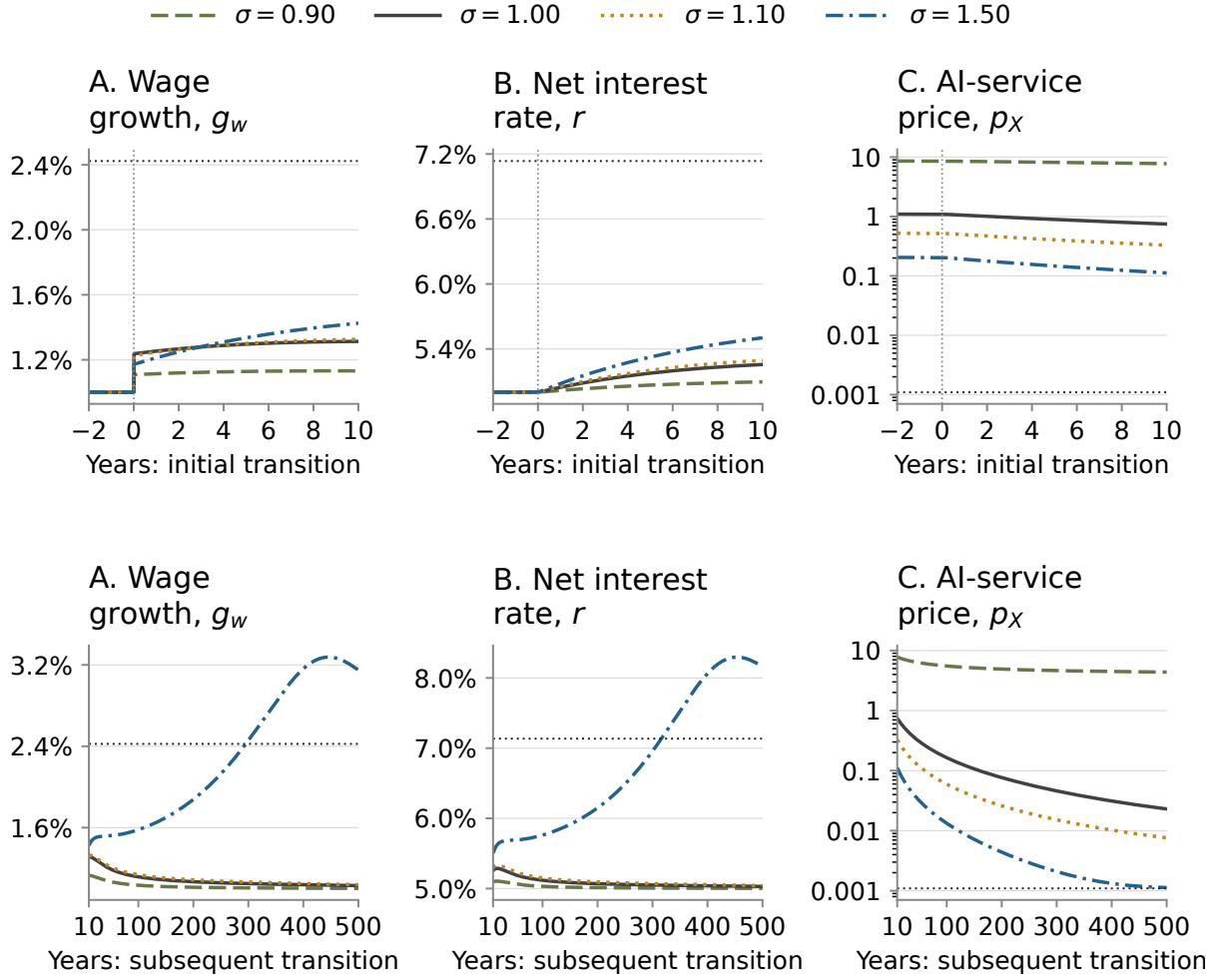


Figure 8: Lower research productivity ($\chi = 1.5$): wage growth, the net interest rate, and the AI-service price. Rates are annual; the price is in final-good units on a logarithmic scale. The vertical line marks RSI activation.

I complete the comparison with Figure 9. Panels A–D report wL/Y , Π/Y , U/Y , and M/Y , all as percentages of output. The upper two rows cover the initial transition and the lower two cover years 10–500, with the same scenario lines. The omitted gross capital share remains 33.0%.

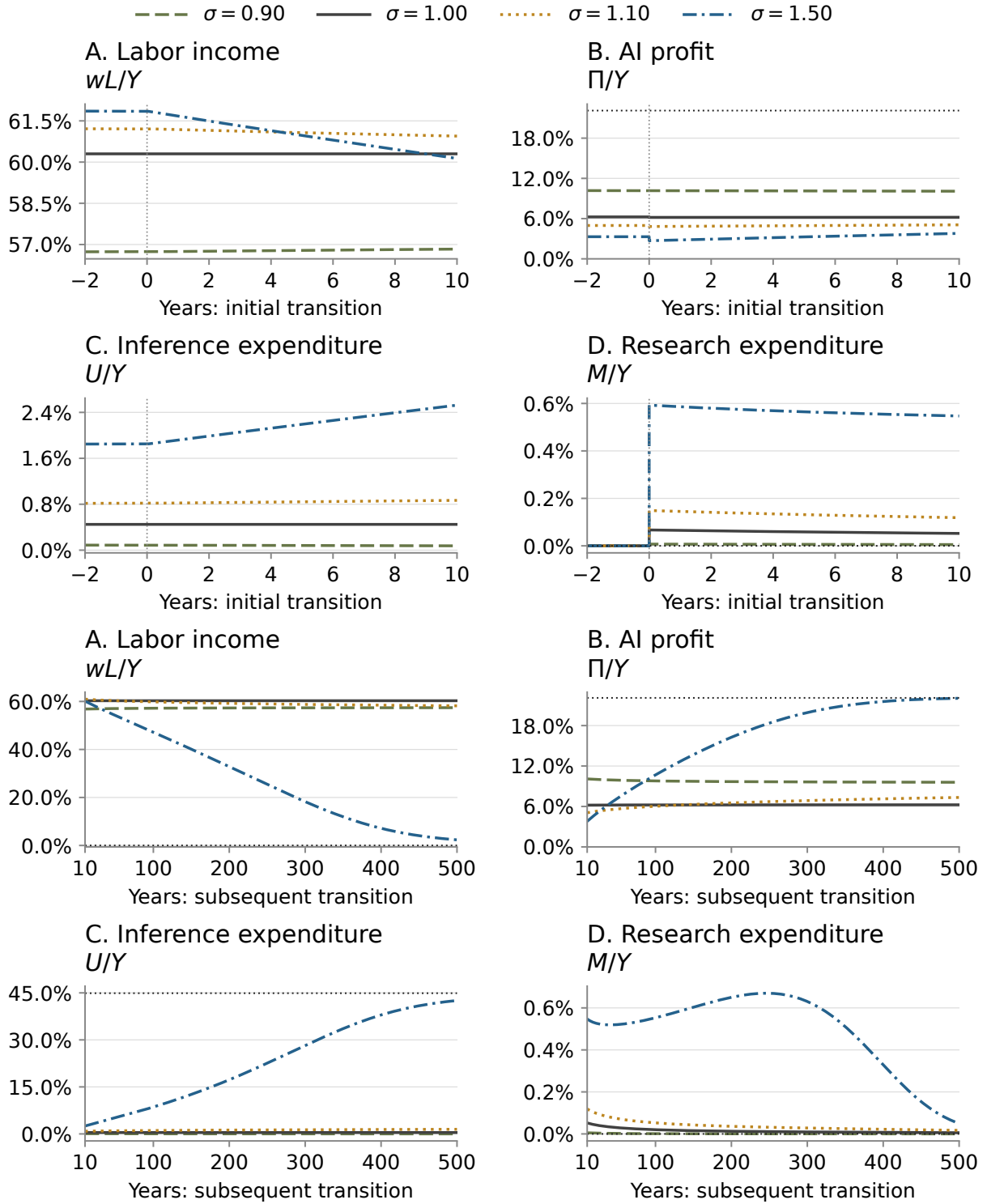


Figure 9: Lower research productivity ($\chi = 1.5$): labor income, AI net profit, inference expenditure, and research expenditure relative to output. The first two rows show the pre-event BGP and years 0–10; the last two show years 10–500. The omitted gross capital share remains $\alpha = 0.33$. Short-run panels A and C use scales fitted to the initial trajectories and omit the long-run reference lines.

In the short run, research absorbs a smaller share of output, leaving more of AI revenue as net profit. At $\sigma = 1.50$, initial research expenditure is 0.6% of output rather than 3.1%, and net profit is 2.7% rather than 0.2%. As in the preceding exercise, the profit reduction finances research, but it is much less pronounced. Inference expenditure and labor’s income share change little in the labor-bottleneck cases. In the AI-dominated case, inference’s share increases and labor’s share declines, but both movements are initially gradual, consistent with the smaller output–wage growth gap.

The longer horizon shows the same eventual redistribution as in Figure 6: labor’s share tends to zero at $\sigma = 1.50$, while inference expenditure and developer profit approach the same positive shares of output. After the initial decline in the research’s share, it rises again before falling toward zero. Expanding AI sales provide an incentive for further research, while proximity to the upper bound eventually limits the efficiency gains it can deliver. The other three economies retain a positive labor share. Lower research productivity changes the adjustment and delays the divergence between regimes without changing their long-run outcomes.

7 Conclusion

This paper endogenizes investment in AI research, which in turn affects the pace of economic growth. Even under RSI, computing requires resources that compete with consumption and accumulation of physical capital. Algorithms and model weights, by contrast, are non-rival: using them does not deplete them. Following [Romer \(1990\)](#), I use monopoly rents to make research investment profitable. In practice, firms compete to develop algorithms and train models. However, as [Korinek and Vipra \(2025\)](#) explain, access to better AI can improve the advantage of a leading developer, so an eventual monopoly does not seem to be far-fetched. The prospect of future monopoly rents can help explain intense competition among firms.

With an upper bound on AI efficiency, permanently faster wage growth requires sufficient substitution between AI and labor and sufficiently high attainable efficiency. In this AI-dominated regime, output and AI revenue per worker grow faster than wages, so labor’s income share vanishes even as wages rise. AI-dominated growth can permanently increase the return on capital, raising questions about wealth concentration not examined here ([Berg et al., 2018](#); [Moll et al., 2022](#)). When labor remains a production bottleneck, the growth rates of output per worker and wages instead return to the exogenous rate of labor-augmenting technological progress. The upper bound on AI efficiency matters for these growth and distributional outcomes: without it, the unit-elasticity case can sustain both growth rates above this exogenous rate while preserving a positive labor share.

RSI is not sufficient for permanently faster output-per-worker growth, nor does such growth require continuing improvements in AI efficiency. Under complementarity, output-per-worker growth cannot permanently exceed the rate of labor-augmenting technological progress, even without an upper bound on AI efficiency. With sufficient substitution and sufficiently high attainable AI efficiency, capital accumulation and expanding inference compute sustain permanently faster growth

in both output per worker and wages, even as AI efficiency converges to its upper bound.

Monopoly pricing also raises the AI-efficiency threshold needed to sustain the higher return associated with AI-dominated growth. This creates a policy tension: monopoly rents reward research, but markups restrict the use of the technology already developed. Removing those rents after innovation can undermine the credibility of the incentives offered beforehand. Patent buyouts could instead reward the developer while opening access to existing AI technology (Kremer, 1998). An extension could study how to value and finance such buyouts while preserving incentives for further RSI.

Simulations show that early growth can look similar across regimes with different long-run outcomes. The question still remains whether human labor will be essential enough for wages to keep pace with the rest of the economy.

A Proofs

Lemma 1 (Static AI-service choice). *For $K, A, L, B > 0$, $0 < \alpha < 1$, $\sigma > 0$, and strictly positive CES weights that sum to one, problem (18) has a unique, strictly positive global maximizer. It is characterized by condition (19) and depends smoothly on the positive variables K, A, L, B , for each fixed σ .*

Proof of Lemma 1. Fix $K, A, L, B > 0$ and let $R(X) = p_X(X)X$ denote AI-service revenue. The inverse-demand elasticity in (15) and differentiation of (10) give

$$R'(X) = p_X(X)(1 - e_X), \quad \frac{ds_X}{d \ln X} = \left(1 - \frac{1}{\sigma}\right) s_X(1 - s_X).$$

Wherever marginal revenue is positive, direct differentiation gives

$$-\frac{XR''(X)}{p_X(X)} = e_X(1 - e_X) + \left(\alpha - \frac{1}{\sigma}\right) \left(1 - \frac{1}{\sigma}\right) s_X(1 - s_X). \quad (64)$$

If $(\alpha - 1/\sigma)(1 - 1/\sigma) \geq 0$, the right-hand side is positive. If this product is negative, the parameter restrictions imply $0 < \alpha < 1/\sigma < 1$, and

$$\begin{aligned} -\frac{XR''(X)}{p_X(X)} &= \frac{1}{\sigma} \left(1 - \frac{1}{\sigma}\right) (1 - s_X)^2 \\ &\quad + \left[\frac{1}{\sigma^2} + 2\alpha - \frac{3\alpha}{\sigma}\right] s_X(1 - s_X) + \alpha(1 - \alpha)s_X^2. \end{aligned}$$

The middle coefficient is positive because

$$\frac{1}{\sigma^2} + 2\alpha - \frac{3\alpha}{\sigma} = \left(\frac{1}{\sigma} - \alpha\right)^2 + \alpha \left(2 - \frac{1}{\sigma} - \alpha\right) > 0.$$

Thus marginal revenue decreases strictly whenever it is positive.

As $X \downarrow 0$, revenue is asymptotically proportional to $X^{1-1/\sigma}$ when $\sigma > 1$, to $X^{(1-\alpha)\omega_X}$ when $\sigma = 1$, and to $X^{1-\alpha}$ when $\sigma < 1$. Every exponent lies between zero and one, so $R'(X) \rightarrow \infty$. As $X \rightarrow \infty$, marginal revenue converges to zero from above if $\sigma \geq 1$; if $\sigma < 1$, it eventually becomes negative. Hence $R'(X) = 1/B$ has exactly one solution. Operating profit increases before this solution and decreases after it. The linear cost X/B dominates revenue as $X \rightarrow \infty$, so this solution is the unique global maximum. Equation (64) also proves the strict second-order condition:

$$e_X(1 - e_X) + \left(\alpha - \frac{1}{\sigma}\right) \left(1 - \frac{1}{\sigma}\right) s_X(1 - s_X) > 0. \quad (65)$$

At the unique solution of $R'(X) = 1/B$, $R''(X) < 0$. The implicit-function theorem therefore makes the maximizing service choice smooth in K, A, L, B . $\square$

Lemma 2 (Optimal research expenditure). *Consider a feasible research trajectory $\{B_t, M_t\}_{t \geq 0}$ for problem (21), starting from B_0 and satisfying the RSI technology (17). Suppose $M_t > 0$ and the present value of net profit is finite. Suppose there exists a continuously differentiable shadow-price trajectory $\{q_t\}_{t \geq 0}$ satisfying the research condition (22), the costate equation (23), and the transversality condition (25).*

Suppose $0 < B_0 < \bar{B} < \infty$ and, for every $t \geq 0$ and every $b \in [B_0, \bar{B})$,

$$\begin{aligned} \pi_t(b) - \pi_t(B_t) + \frac{1-\eta}{\eta} M_t \left[\left(\frac{b}{B_t}\right)^{\eta/(1-\eta)} - 1 \right] \\ \leq \frac{\bar{B} - B_t}{B_t} (U_t + M_t) \log \left(\frac{\bar{B} - B_t}{\bar{B} - b} \right), \end{aligned} \quad (66)$$

where $\pi_t(b)$ is operating profit as defined in equation (18). Then $\{M_t\}_{t \geq 0}$ solves the developer's problem (21): no other feasible research trajectory starting from the same B_0 yields a higher present value of net profit.

Proof of Lemma 2. I first transform AI efficiency to compare the candidate with alternative research trajectories. Define

$$R_B(B) = -\bar{B} \log \left(1 - \frac{B}{\bar{B}} \right), \quad B(R_B) = \bar{B} \left(1 - e^{-R_B/\bar{B}} \right). \quad (67)$$

It maps $B \in [B_0, \bar{B})$ into $R_B \in [R_{B,0}, \infty)$, where $R_{B,t} = R_B(B_t)$ and $R_{B,0} = R_B(B_0)$. Since $R'_B(B) = 1/\psi(B)$, this transformation removes the factor $\psi(B)$ from the RSI technology (17):

$$\dot{R}_B = \chi[B(R_B)M]^\eta. \quad (68)$$

The corresponding current-value shadow price is $q_{R,t} = q_t \psi(B_t)$, which is positive by the research condition (22). At each date t , hold $q_{R,t}$ at its value along the candidate trajectory and define the

maximized Hamiltonian for an alternative state $\tilde{R}_B$:

$$\hat{\mathcal{H}}_t(\tilde{R}_B) \equiv \max_{\tilde{M} \geq 0} \left\{ \pi_t(B(\tilde{R}_B)) - \tilde{M} + q_{R,t} \chi[B(\tilde{R}_B)\tilde{M}]^\eta \right\}.$$

Because $0 < \eta < 1$, the expression being maximized is strictly concave in research expenditure. Using the candidate's research condition (22), the maximizing expenditure at $\tilde{R}_B$ is

$$M_t \left[\frac{B(\tilde{R}_B)}{B_t} \right]^{\eta/(1-\eta)},$$

where M_t is the candidate's research expenditure. Substitution gives

$$\hat{\mathcal{H}}_t(\tilde{R}_B) = \pi_t(B(\tilde{R}_B)) + \frac{1-\eta}{\eta} M_t \left[\frac{B(\tilde{R}_B)}{B_t} \right]^{\eta/(1-\eta)}. \quad (69)$$

Differentiating the maximized Hamiltonian with respect to its state argument and using the envelope condition (20) gives

$$\hat{\mathcal{H}}'_t(R_{B,t}) = \frac{\psi(B_t)}{B_t} (U_t + M_t). \quad (70)$$

Because

$$R_B(b) - R_{B,t} = \bar{B} \log \left(\frac{\bar{B} - B_t}{\bar{B} - b} \right),$$

condition (66) is equivalent to the following inequality for every $\tilde{R}_B \geq R_{B,0}$:

$$\hat{\mathcal{H}}_t(\tilde{R}_B) \leq \hat{\mathcal{H}}_t(R_{B,t}) + \hat{\mathcal{H}}'_t(R_{B,t})(\tilde{R}_B - R_{B,t}).$$

Let $D_t = \exp(-\int_0^t r_s ds)$ be the discount factor and $\mu_t = D_t q_{R,t}$ the present-value shadow price of the transformed state R_B . The research condition (22) and the RSI technology (17) imply $q_t \eta \dot{B}_t = M_t$. Differentiating $q_{R,t} = q_t \psi(B_t)$ and using the costate equation (23) therefore gives

$$\dot{q}_{R,t} = r_t q_{R,t} - \frac{\psi(B_t)}{B_t} (U_t + M_t), \quad \dot{\mu}_t = -D_t \hat{\mathcal{H}}'_t(R_{B,t}).$$

Consider any feasible alternative $\{\tilde{B}_t, \tilde{M}_t\}_{t \geq 0}$ starting from the same B_0 , and write $\tilde{R}_{B,t} = R_B(\tilde{B}_t)$. Here feasible expenditure is nonnegative and integrable on every finite interval, with a locally absolutely continuous state satisfying the RSI technology almost everywhere. Let

$$J_T = \int_0^T D_t [\pi_t(B_t) - M_t] dt$$

denote discounted net profit through date T under the candidate, and define $\tilde{J}_T$ analogously for the alternative. At each date, the alternative's Hamiltonian cannot exceed $\hat{\mathcal{H}}_t(\tilde{R}_{B,t})$, whereas the candidate attains $\hat{\mathcal{H}}_t(R_{B,t})$. Applying condition (66) and the transformed costate equation yields,

for almost every t ,

$$\begin{aligned} D_t \left[\pi_t(\tilde{B}_t) - \tilde{M}_t - \pi_t(B_t) + M_t \right] \\ \leq D_t \hat{\mathcal{H}}'_t(R_{B,t})(\tilde{R}_{B,t} - R_{B,t}) - \mu_t(\dot{\tilde{R}}_{B,t} - \dot{R}_{B,t}) \\ = -\frac{d}{dt} \left[\mu_t(\tilde{R}_{B,t} - R_{B,t}) \right]. \end{aligned}$$

Integrating from zero to T and using the common initial state gives

$$\tilde{J}_T - J_T \leq -\mu_T(\tilde{R}_{B,T} - R_{B,T}) \leq \mu_T R_{B,T},$$

where the second inequality uses $\mu_T > 0$ and $\tilde{R}_{B,T} \geq 0$. Since

$$\psi(B_t)R_{B,t} = -(\bar{B} - B_t) \log \left(1 - \frac{B_t}{\bar{B}} \right) \leq B_t,$$

the developer's transversality condition (25) gives

$$0 \leq \mu_T R_{B,T} = D_T q_T \psi(B_T) R_{B,T} \leq D_T q_T B_T \longrightarrow 0.$$

The candidate's present value is finite by assumption. Hence

$$\limsup_{T \rightarrow \infty} \tilde{J}_T \leq \lim_{T \rightarrow \infty} J_T < \infty,$$

so no feasible alternative yields a higher present value of net profit. The argument requires the tangent to the maximized Hamiltonian at the candidate state to lie above its value at every alternative state. Global concavity is sufficient for this inequality, but is not required. $\square$

Lemma 3 (Sufficient conditions for Hamiltonian concavity). *Consider the maximized Hamiltonian in equation (69) for a feasible trajectory satisfying the research condition (22), with $M_t > 0$ and $0 < B_0 < \bar{B} < \infty$. Let $X_t^*(b)$ denote the optimal AI-service choice in problem (18) at efficiency b , and let $\varepsilon_{B,t}(b) = \partial \ln X_t^*(b) / \partial \ln b$ denote its elasticity with respect to AI efficiency. Suppose that, for every $t \geq 0$ and $b \in [B_0, \bar{B})$,*

$$\psi(b)[\varepsilon_{B,t}(b) - 2] \leq \frac{b}{\bar{B}}, \quad (71)$$

and

$$\frac{B_0}{\bar{B}} \geq \max \left\{ 0, \frac{2\eta - 1}{\eta} \right\}. \quad (72)$$

Then the maximized Hamiltonian is concave in the transformed state $R_B \in [R_B(B_0), \infty)$ at every date. Consequently, the support condition (66) in Lemma 2 holds.

Proof of Lemma 3. Fix a date t . Lemma 1 ensures that $X_t^*(b)$ is unique and smooth. Let $s_{X,t}(b)$ and $e_{X,t}(b)$ be the corresponding elasticities defined in equations (10) and (15). Implicit differentiation

of the monopoly condition (19) gives

$$\varepsilon_{B,t}(b) \equiv \frac{\partial \ln X_t^*(b)}{\partial \ln b} = \frac{1 - e_{X,t}(b)}{e_{X,t}(b)[1 - e_{X,t}(b)] + (\alpha - \sigma^{-1})(1 - \sigma^{-1})s_{X,t}(b)[1 - s_{X,t}(b)]}. \quad (73)$$

The denominator is strictly positive by equation (65). The envelope condition (20) gives $\pi'_t(b) = X_t^*(b)/b^2$, so

$$\pi''_t(b) = \frac{X_t^*(b)}{b^3} [\varepsilon_{B,t}(b) - 2]. \quad (74)$$

The transformation (67) satisfies $B'(R_B) = \psi(b)$ and $B''(R_B) = -\psi(b)/\bar{B}$, where $b = B(R_B)$. The chain rule therefore gives

$$\frac{d^2 \pi_t(B(R_B))}{dR_B^2} = \frac{X_t^*(b)\psi(b)}{b^3} \left[\psi(b)(\varepsilon_{B,t}(b) - 2) - \frac{b}{\bar{B}} \right]. \quad (75)$$

Condition (71) makes this derivative nonpositive for every $b \in [B_0, \bar{B})$.

The research term in (69) is a positive multiple of $B(R_B)^{\eta/(1-\eta)}$. Its curvature is

$$\frac{d^2 B(R_B)^{\eta/(1-\eta)}}{dR_B^2} = \frac{\eta}{1-\eta} b^{\eta/(1-\eta)-2} \psi(b) \left[\frac{2\eta-1}{1-\eta} \psi(b) - \frac{b}{\bar{B}} \right]. \quad (76)$$

The bracket is nonpositive exactly when $b/\bar{B} \geq (2\eta-1)/\eta$. Since $b \geq B_0$, condition (72) makes the research term concave as well. The maximized Hamiltonian is therefore concave, so its tangent at the candidate state is a global upper bound. As shown in the proof of Lemma 2, this is condition (66).

If $\eta \leq 1/2$, condition (72) holds automatically. But for every fixed $0 < \eta < 1$ its right-hand side is strictly below one, so it also holds without this restriction when initial efficiency is sufficiently close to the upper bound. These curvature tests are sufficient, not necessary: they can fail even when the Hamiltonian has the required supporting tangent at the candidate state. $\square$

Proof of Proposition 1. The proof has three tasks. First, Lemma 1 gives a unique, smooth, globally optimal within-date service choice at given states. I combine this choice with the other equilibrium equations to identify candidate long-run values and verify their admissibility, including positive consumption. Finding these values does not yet establish the existence of a trajectory approaching them.

Second, I write the exact dynamic equations in coordinates with finite candidate limits. The linearization identifies the stable directions; I also check that they can accommodate every sufficiently nearby initial state. Counting stable directions alone would not establish this property. The stable-manifold and inverse-function theorems then select the initial consumption and shadow price for those states. They establish convergent solutions of the nonlinear equations for all future time, not merely solutions of their linear approximation.

Third, I reconstruct the original variables and verify feasibility, market clearing, and both transversality conditions. Household concavity and Lemma 2 establish optimality. For the developer,

I verify the curvature conditions of Lemma 3 at every AI-efficiency level reachable under alternative research policies from the initial efficiency. The upper bound makes this possible for initial efficiency sufficiently close to it. These final checks turn the constructed solutions of the first-order conditions into equilibria. The result is local in initial states, not in the time horizon.

Step 1. Feasible efficiency and candidate long-run values.

Let $D = \bar{B} - B$. The state equation with an upper bound gives

$$\frac{\dot{D}}{D} = -\frac{\chi}{\bar{B}}(BM)^\eta.$$

Integration gives

$$\bar{B} - B_t = (\bar{B} - B_0) \exp \left\{ -\frac{\chi}{\bar{B}} \int_0^t (B_s M_s)^\eta ds \right\} > 0. \quad (77)$$

Local integrability makes the exponent finite at every finite date, so $D_t > 0$.

For the candidate limits, impose $B = \bar{B}$ only in the normalized boundary equations. Actual trajectories will satisfy $B < \bar{B}$ at every finite date.

Write $\theta = (1 - \alpha)/\alpha$. For $\sigma \neq 1$, let $\varphi = (\sigma - 1)/\sigma$ and define

$$m(s) = 1 - \frac{1-s}{\sigma} - \alpha s, \quad \mathcal{H}_\sigma(s) = \left[(1 - \alpha)s m(s) \left(\frac{\omega_X}{s} \right)^{1/\varphi} \right]^\theta.$$

Here $m(s) = 1 - e_X$ is the marginal-revenue factor, not research compute. The static monopoly optimum has $m(s_X) > 0$ by Lemma 1. The CES share, monopoly condition, and production function, equations (28d), (28i), and (28c), imply

$$\frac{X}{AN} = \left[\frac{\omega_L s_X}{\omega_X (1 - s_X)} \right]^{1/\varphi}, \quad \frac{U}{Y} = (1 - \alpha)s_X m(s_X), \quad (78)$$

$$\frac{Y}{K} = B^\theta \mathcal{H}_\sigma(s_X). \quad (79)$$

The identities $Z = X(\omega_X/s_X)^{1/\varphi}$ and $X = B(1 - \alpha)s_X m(s_X)Y$ give the output–capital ratio in (79).

If $0 < \sigma < 1$, positive marginal revenue requires

$$s > s_{\min} \equiv \frac{1 - \sigma}{1 - \alpha\sigma}.$$

On $(s_{\min}, 1)$, both $m(s)$ and $s^{-1/(\sigma-1)}$ increase strictly. Since

$$\mathcal{H}_\sigma(s)^{1/\theta} = (1 - \alpha)\omega_X^{1/\varphi} m(s)s^{-1/(\sigma-1)},$$

the share map increases from zero to $\mathcal{H}_\sigma(1) = [(1 - \alpha)^2 \omega_X^{1/\varphi}]^\theta$. Thus the equation

$$\frac{\rho + \gamma + \delta}{\alpha} = \bar{B}^\theta \mathcal{H}_\sigma(\bar{s}_X) \quad (80)$$

has exactly one interior solution if and only if $\bar{B} > \mathcal{B}(\sigma)$.

If $\sigma > 1$, the share map decreases strictly on $(0, 1)$. To verify the sign, put $a = \sigma^{-1} - \alpha$, so $m(s) = \varphi + as$. Its logarithmic derivative has the sign of $a(\sigma - 2)s - \varphi$. If $a(\sigma - 2) \leq 0$ this is negative. If $\sigma > 2$ and $a > 0$, $a(\sigma - 2) < (\sigma - 2)/\sigma < \varphi$; if $1 < \sigma < 2$ and $a < 0$, $a(\sigma - 2) < (1 - \sigma^{-1})(2 - \sigma) < \varphi$. The map therefore decreases from infinity to $\mathcal{H}_\sigma(1)$. Equation (80) has exactly one interior solution if and only if $\bar{B} < \mathcal{B}(\sigma)$. Since $\mathcal{R}(B) = \alpha B^\theta \mathcal{H}_\sigma(1) - \delta$, these calculations also derive (35)–(36). The interior-share condition is therefore $\mathcal{R}(\bar{B}) > \rho + \gamma$ for $\sigma < 1$, but $\mathcal{R}(\bar{B}) < \rho + \gamma$ for $\sigma > 1$.

For either interior-share case, let x, y, u, k, c denote X, Y, U, K, C divided by AN , respectively. Construct the normalized terminal values sequentially:

$$\begin{aligned} \bar{x} &= \left[\frac{\omega_L \bar{s}_X}{\omega_X (1 - \bar{s}_X)} \right]^{1/\varphi}, & \bar{u} &= \frac{\bar{x}}{\bar{B}}, \\ \bar{y} &= \frac{\bar{u}}{(1 - \alpha) \bar{s}_X m(\bar{s}_X)}, & \bar{k} &= \frac{\alpha \bar{y}}{\rho + \gamma + \delta}, \\ \bar{c} &= \bar{y} - \bar{u} - (\delta + n + \gamma) \bar{k}, & \bar{r} &= \rho + \gamma. \end{aligned} \quad (81)$$

At $\sigma = 1$, define $\beta = (1 - \alpha)\omega_X$. The exact static solution is

$$y = \left[(\beta^2 B)^\beta k^\alpha \right]^{1/(1-\beta)}, \quad u = \beta^2 y, \quad x = Bu.$$

The equation $y/k = (\rho + \gamma + \delta)/\alpha$ has exactly one positive solution for every finite $\bar{B} > 0$:

$$\bar{k} = \left[(\beta^2 \bar{B})^\beta \left(\frac{\rho + \gamma + \delta}{\alpha} \right)^{\beta-1} \right]^{1/(1-\alpha-\beta)}. \quad (82)$$

Set $\bar{y} = [(\rho + \gamma + \delta)/\alpha] \bar{k}$, $\bar{u} = \beta^2 \bar{y}$, and $\bar{x} = \bar{B} \bar{u}$, with $\bar{c}, \bar{r}$ as in (81).

Consumption positivity does not require an additional restriction. Positive marginal revenue implies $0 < m(\bar{s}_X) < 1$. Since $0 < \bar{s}_X < 1$, in all three cases

$$\frac{\bar{u}}{\bar{y}} = (1 - \alpha) \bar{s}_X m(\bar{s}_X) < 1 - \alpha, \quad \frac{(\delta + n + \gamma) \bar{k}}{\bar{y}} = \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} < \alpha.$$

It follows that $\bar{c}/\bar{y} > 0$.

Step 2. The three positive-labor-share cases.

The cases in Proposition 1(i), 1(ii), and 1(iii) share one normalization. Divide capital, consumption, and the AI-efficiency shadow price by effective labor, and multiply the remaining efficiency gap by effective labor:

$$k = \frac{K}{AN}, \quad c = \frac{C}{AN}, \quad d = (\bar{B} - B)AN, \quad \tilde{q} = \frac{q}{AN}, \quad \tau = (AN)^{-1}. \quad (83)$$

Dividing by AN removes effective-labor growth. Multiplying the shrinking gap by AN allows it to converge to a positive value: $\dot{d}/d = n + \gamma - \dot{B}/(\bar{B} - B)$. The variable d measures the scaled gap,

not its rate of closure. The constants $\bar{k}, \bar{d}$ are the limiting values of k, d ; they are determined by the model, not additional parameters. Also $\dot{\tau} = -(n + \gamma)\tau$, so $\tau \rightarrow 0$. These coordinates leave the model unchanged: $K = k/\tau$ and $B = \bar{B} - d\tau$ at every finite date.

Given k, B , solve the unique static monopoly problem for y, x, u , and set $r = \alpha y/k - \delta$. Define $a = \dot{B}/(\bar{B} - B)$, the proportional rate of closure of the AI-efficiency gap. The research condition gives

$$B = \bar{B} - d\tau, \quad M = \left[\tilde{q} d \chi \eta \frac{B^\eta}{\bar{B}} \right]^{1/(1-\eta)}, \quad a = \chi \frac{B^\eta M^\eta}{\bar{B}}.$$

Substituting (28k) and (29) gives the exact dynamic equations

$$\frac{\dot{k}}{k} = \frac{y - c - u - M\tau}{k} - \delta - n - \gamma, \quad (84a)$$

$$\frac{\dot{c}}{c} = r - \rho - \gamma, \quad (84b)$$

$$\frac{\dot{d}}{d} = n + \gamma - a, \quad (84c)$$

$$\frac{\dot{\tilde{q}}}{\tilde{q}} = r - \frac{x}{\tilde{q} B^2} - \eta \frac{a d \tau}{B} + a - n - \gamma, \quad (84d)$$

$$\dot{\tau} = -(n + \gamma)\tau. \quad (84e)$$

In particular, population growth cancels from the normalized consumption equation; it is not subtracted a second time.

Complete the terminal construction by setting

$$\bar{M} = \left[\frac{(n + \gamma) \bar{B}^{1-\eta}}{\chi} \right]^{1/\eta}, \quad \bar{\tilde{q}} = \frac{\bar{x}}{\bar{r} \bar{B}^2}, \quad \bar{d} = \frac{\bar{M}}{\eta(n + \gamma) \bar{\tilde{q}}}. \quad (85)$$

All are positive. Direct substitution shows that $(\bar{k}, \bar{c}, \bar{d}, \bar{\tilde{q}}, 0)$ is a fixed point of (84). This boundary point is a device for constructing paths with $\tau > 0$ and $B < \bar{B}$, not an admissible date-zero state itself.

I next verify saddle-path stability. Write $\mu = \eta/(1 - \eta)$, $r_k = \partial r / \partial \log k$, and $a_k = \partial[(y - u)/k] / \partial \log k + \bar{c}/\bar{k}$, evaluated at the fixed point. In $(\log k, \log c, \log d, \log \tilde{q}, \tau)$ coordinates the Jacobian has two diagonal blocks: one for capital and consumption, and one for the scaled efficiency gap and normalized shadow price,

$$J_R = \begin{pmatrix} a_k & -\bar{c}/\bar{k} \\ r_k & 0 \end{pmatrix}, \quad J_B = \begin{pmatrix} -(n + \gamma)\mu & -(n + \gamma)\mu \\ (n + \gamma)\mu & \bar{r} + (n + \gamma)\mu \end{pmatrix}, \quad (86)$$

and the additional eigenvalue $-(n + \gamma)$. There can be forcing from τ to both blocks and from (k, c) to $(d, \tilde{q})$, but not in the reverse directions at the fixed point.

To check $r_k < 0$, let $s = s_X$, $u_\sigma = 1/\sigma$, $e = e_X$, and let $D(s)$ be the positive curvature expression

in (64). Implicit differentiation of the monopoly condition gives

$$\frac{\partial \log X}{\partial \log K} = \frac{\alpha(1-e)}{D(s)}, \quad r_k = (r + \delta)(1 - \alpha) \left[\frac{\alpha s(1-e)}{D(s)} - 1 \right].$$

Moreover,

$$D(s) - \alpha s(1-e) = (1-s)\{u_\sigma(1-e) + (\alpha - u_\sigma)(1-u_\sigma)s\} > 0.$$

The sign is immediate when $u_\sigma \geq 1$ or $u_\sigma \leq \alpha$. For $\alpha < u_\sigma < 1$, the braces are linear in s , with positive endpoint values $u_\sigma(1-u_\sigma)$ and $u_\sigma^2 + \alpha(1-2u_\sigma)$. The latter is positive if $u_\sigma \leq 1/2$; otherwise $\alpha < u_\sigma$ implies it exceeds $u_\sigma(1-u_\sigma) > 0$. Thus $r_k < 0$ in every interior-share case.

Both blocks have negative determinants:

$$\det J_R = (\bar{c}/\bar{k})r_k < 0, \quad \det J_B = -(n + \gamma)\mu\bar{r} < 0.$$

There are therefore three stable and two unstable eigenvalues. The stable direction in J_R has a nonzero k component, and the stable direction in J_B has a nonzero d component. To verify that their number suffices to select the jump variables, consider a vector in the stable subspace whose state components (k, d, τ) are all zero. Its linearized τ path is zero. The unforced capital–consumption block must then follow its stable direction; its zero initial k component forces its c component to be zero as well. With that forcing absent, the same argument in the research block forces $\tilde{q} = 0$. The projection of the three-dimensional stable subspace onto the three predetermined coordinates is therefore injective and hence invertible. This argument also covers repeated stable eigenvalues.

Lemma 1 makes the static solution smooth. The normalized vector field extends smoothly through $\tau = 0$, since $B = \bar{B} - d\tau$ remains positive nearby. The stable-manifold theorem and the inverse-function theorem therefore give a local graph $(c, \tilde{q})$ over (k, d, τ) ; see Dyatlov (2018). Choose a small neighborhood of the full fixed point and then a smaller initial-state neighborhood $\mathcal{U}_L$ of $(\bar{k}, \bar{d}, 0)$. The size of this neighborhood depends on the parameters and the case under consideration. In original initial stocks the restriction is

$$(k_0, d_0, \tau_0) = \left(\frac{K_0}{A_0 N_0}, (\bar{B} - B_0)A_0 N_0, \frac{1}{A_0 N_0} \right) \in \mathcal{U}_L. \quad (87)$$

Closeness is required in all three coordinates; B_0 close to $\bar{B}$ alone is not sufficient. The point with $\tau = 0$ is a long-run limit, not a finite-date initial state. Every physical initial state satisfying (87) has a solution on this graph that stays in the full neighborhood for all future time and converges to the fixed point. It is unique among solutions with those properties. Positivity of $\tau, d, c, \tilde{q}, k$ and the original RSI technology ensure $0 < B_t < \bar{B}$ and $M_t > 0$ at every finite date. Reconstruction gives the original initial stocks, the equilibrium equations, and the limiting rates in (37)–(39). Indeed, the normalized vector field tends to zero at its fixed point, so the time derivatives of its smooth static ratios tend to zero as well. This step, not convergence of ratios alone, establishes convergence of the growth rates. Optimality and transversality are verified below.

Step 3. The AI-dominated case.

Let $\sigma > 1$ and $\bar{B} > \mathcal{B}(\sigma)$. Here $\varphi = (\sigma - 1)/\sigma \in (0, 1)$. I seek an equilibrium in which capital grows faster than effective labor. Along such a trajectory, $K/(AN)$ would diverge, so I normalize consumption and the shadow value by capital and scale the remaining AI-efficiency gap by capital:

$$h = (AN/K)^\varphi, \quad d = (\bar{B} - B)K, \quad c = C/K, \quad v = q/K, \quad \tau = (AN)^{-1}.$$

Within this construction, y, x, u denote $Y/K, X/K, U/K$, respectively. The constant $\bar{d}$ now denotes the limit of $(\bar{B} - B)K$, not of $(\bar{B} - B)AN$; no original state or control is redefined. Then

$$B = \bar{B} - d\tau h^{1/\varphi}, \quad M = [vd\chi\eta B^\eta/\bar{B}]^{1/(1-\eta)}, \quad a = \chi B^\eta M^\eta/\bar{B}, \quad r = \alpha y - \delta.$$

Capital's growth rate is $g_K = y - c - u - M\tau h^{1/\varphi} - \delta$. The exact equations are

$$\dot{h} = \varphi h(n + \gamma - g_K), \tag{88a}$$

$$\dot{d}/d = g_K - a, \tag{88b}$$

$$\dot{c}/c = n + r - \rho - g_K, \tag{88c}$$

$$\dot{v}/v = r - x/(vB^2) - \eta ad\tau h^{1/\varphi}/B + a - g_K, \tag{88d}$$

$$\dot{\tau} = -(n + \gamma)\tau. \tag{88e}$$

The CES term divided by capital is $[\omega_L h + \omega_X x^\varphi]^{1/\varphi}$. At $h = 0$ the monopoly equation has the strictly positive curvature $D(1) = \alpha(1 - \alpha)$. Its implicit static solution is therefore smooth in (h, B) near $(0, \bar{B})$. For the remaining terms, replacing $h^{1/\varphi}$ by $\max\{h, 0\}^{1/\varphi}$ outside the physical domain gives a C^1 extension because $1/\varphi > 1$. The C^1 stable-manifold theorem applies; see [Feldman \(2021, p. 7\)](#). The extension is only a mathematical device and does not change any equation on $h \geq 0$.

The limiting interest and aggregate growth rates are

$$\bar{r} = \mathcal{R}(\bar{B}), \quad \bar{g} = n + \bar{r} - \rho > n + \gamma. \tag{89}$$

At $h = \tau = 0$, the terminal constants are

$$\begin{aligned} \bar{y} &= \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha}, & \bar{u} &= (1 - \alpha)^2 \bar{y}, & \bar{x} &= \bar{B} \bar{u}, \\ \bar{r} &= \mathcal{R}(\bar{B}), & \bar{g} &= n + \bar{r} - \rho, & \bar{c} &= \alpha(1 - \alpha) \bar{y} + \rho - n > 0, \\ \bar{v} &= \bar{x}/(\bar{r} \bar{B}^2), & \bar{M} &= \left[\bar{g} \bar{B}^{1-\eta}/\chi \right]^{1/\eta}, & \bar{d} &= \bar{M}/(\eta \bar{g} \bar{v}). \end{aligned} \tag{90}$$

The threshold implies $\bar{g} > n + \gamma > 0$ and $\bar{r} > \rho + \gamma > 0$, so all constants are positive. Substitution verifies the fixed point $(0, \bar{d}, \bar{c}, \bar{v}, 0)$.

The h and τ eigenvalues are $\varphi(n + \gamma - \bar{g}) < 0$ and $-(n + \gamma) < 0$. The first equals $\varphi[\rho + \gamma - \mathcal{R}(\bar{B})]$. It becomes zero at $\mathcal{R}(\bar{B}) = \rho + \gamma$, which is why this stable-manifold argument does not cover the

threshold. The consumption-ratio eigenvalue is $\bar{c} > 0$. The remaining block in $(\log d, \log v)$ is

$$\begin{pmatrix} -\bar{g}\mu & -\bar{g}\mu \\ \bar{g}\mu & \bar{r} + \bar{g}\mu \end{pmatrix}, \quad \det = -\bar{g}\mu\bar{r} < 0.$$

The linearized system is triangular in the order $(h, \tau), c, (d, v)$. A stable vector with zero state components (h, d, τ) has zero h, τ paths; its consumption component must be zero because the unforced consumption equation is unstable. The research block then has zero d component and hence zero v component, as in the previous construction. Thus the three-dimensional stable subspace projects invertibly onto (h, d, τ) . The same graph argument gives a neighborhood $\mathcal{U}_A$ of $(0, \bar{d}, 0)$ such that the initial-state condition is

$$\left[\left(\frac{A_0 N_0}{K_0} \right)^\varphi, (\bar{B} - B_0)K_0, \frac{1}{A_0 N_0} \right] \in \mathcal{U}_A. \quad (91)$$

It constructs a convergent trajectory for every admissible initial state in this neighborhood, with a unique nearby convergent choice of C_0, q_0 and positive quantities at every finite date.

The normalized vector field tends to zero, so the time derivatives of the smooth positive ratios Y/K , X/K , and C/K tend to zero. Consequently $g_Y, g_X, g_C, g_K \rightarrow \bar{g}$ and $s_X \rightarrow 1$. Log differentiation of the CES share in equation (28d) and the wage equation (28g) gives the exact identity

$$g_w = g_Y - n + \frac{\sigma - 1}{\sigma} s_X (n + \gamma - g_X).$$

Taking limits yields

$$g_w \rightarrow \bar{g} - n + \frac{\sigma - 1}{\sigma} (n + \gamma - \bar{g}) = \gamma + \frac{\bar{r} - \rho - \gamma}{\sigma}.$$

This proves the growth and distribution limits in Proposition 1(iv).

Step 4. Optimality and transversality.

In either construction let $\bar{g}$ denote the terminal aggregate growth rate: $n + \gamma$ in the positive labor-share case and $n + \bar{r} - \rho$ in the AI-dominated case. The constructions imply

$$a \rightarrow \bar{g}, \quad M \rightarrow \left[\frac{\bar{g} \bar{B}^{1-\eta}}{\chi} \right]^{1/\eta} > 0, \quad g_q \rightarrow \bar{g}, \quad g_B \rightarrow 0, \quad M/Y \rightarrow 0.$$

Both transversality expressions in Definition 1 have limiting logarithmic growth

$$-\rho + n + g_K - g_C \rightarrow n - \rho < 0, \quad -r + g_q + g_B \rightarrow \bar{g} - \bar{r} = n - \rho < 0. \quad (92)$$

They therefore converge to zero, rather than merely remaining bounded.

Household utility is finite because $\log(C/N)$ grows at most linearly and $\rho > n$, as imposed in the household problem. Log utility is concave, and the household budget is linear at given prices and distributed profits. More explicitly, let $\lambda_t = e^{-\rho t} N_t / C_t$, so $\dot{\lambda}_t = -r_t \lambda_t$ by (5). Concavity of log

utility and equal initial capital give, for any admissible alternative $(\tilde{C}, \tilde{K})$,

$$\int_0^T e^{-\rho t} N_t \log \frac{\tilde{C}_t}{C_t} dt \leq \int_0^T \lambda_t (\tilde{C}_t - C_t) dt = -\lambda_T (\tilde{K}_T - K_T) \leq \lambda_T K_T \rightarrow 0.$$

Here $\tilde{K}_T \geq 0$ and the last limit is the household TVC. Final-good optimality follows from concavity and constant returns. The pointwise monopoly optimum is global by Lemma 1.

For the dynamic developer, feasibility and first-order conditions still require a sufficiency argument. Let $S_t = A_t N_t$ in the positive-share construction and $S_t = K_t$ in the AI-dominated construction, and let $D_t = \exp(-\int_0^t r_s ds)$. Homogeneity of the static problem and compactness of the normalized state neighborhood give a constant C_π such that $0 \leq \pi_t(b) \leq C_\pi S_t$ for every date and every $b \in [B_0, \bar{B}]$. Since $d \log(D_t S_t)/dt \rightarrow \bar{g} - \bar{r} = n - \rho < 0$, $\int_0^\infty D_t S_t dt < \infty$. Thus discounted operating profit is uniformly bounded above across all alternative research policies. The candidate's present value of net profit is finite, because its research compute is bounded and $r_t \rightarrow \bar{r} > 0$. An alternative with infinite discounted research expenditure has payoff $-\infty$ and cannot improve on it.

It remains to verify the curvature conditions of Lemma 3, (71) and (72) for every date and every alternative AI-efficiency level in $[B_0, \bar{B}]$, not just on the selected path. First fix a compact normalized-state neighborhood and an interval of alternative AI-efficiency levels $[b_*, \bar{B}]$, with $0 < b_* < \bar{B}$. In the positive labor-share construction, $K/(AN)$ stays in a compact neighborhood of $\bar{k} > 0$. Smoothness of the static monopoly solution bounds $\varepsilon_{B,t}(b)$ uniformly for these states and for AI-efficiency levels near $\bar{B}$. In the AI-dominated construction, h stays near zero, and the same uniform bound follows from the smooth static limit $\varepsilon_{B,t}(b) \rightarrow 1/\alpha$. Thus in either case there is a finite constant E with $\varepsilon_{B,t}(b) - 2 \leq E$ throughout that state domain. This bound is chosen before shrinking the initial-state neighborhood. Because $\mu = \eta/(1 - \eta)$ is finite for each $0 < \eta < 1$, choose $b_{**} \in [b_*, \bar{B})$ sufficiently close to $\bar{B}$ that

$$(1 - b_{**}/\bar{B}) \max\{E, \mu - 1, 0\} < b_{**}/\bar{B}.$$

Shrink the open neighborhood $\mathcal{U}_L$ or $\mathcal{U}_A$ so that $B_0 > b_{**}$. This is possible because $B_0 = \bar{B} - d_0 \tau_0$ in the first construction and $B_0 = \bar{B} - d_0 \tau_0 h_0^{1/\varphi}$ in the second. Every feasible research policy has $b \geq B_0$, so the two curvature conditions hold throughout the entire alternative-state domain at every date. Lemma 3 therefore gives the support inequality (66). Lemma 2 and the verified TVC then establish global developer optimality without imposing $\eta \leq 1/2$. The curvature conditions follow from the primitives and the chosen initial-state neighborhood.

All quantities remain finite at every finite date, and market clearing and the original initial conditions hold by reconstruction. The paths are therefore equilibria, not merely solutions of the canonical equations. The result is local only in its set of initial stocks. For $\sigma > 1$, at $\bar{B} = \mathcal{B}(\sigma)$ the interior-share terminal point reaches the boundary and a stable eigenvalue in the boundary normalization becomes zero; this proof does not apply. For $\sigma < 1$ and $\bar{B} \leq \mathcal{B}(\sigma)$, it does not rule out equilibria with different long-run ratios. Nor does it imply existence at arbitrary initial stocks or in the uncapped limit.

Step 5. Initial-state sets and the four conclusions.

The preimage of each open neighborhood under its continuous initial-state map, restricted to $A_0, N_0, K_0 > 0$ and $0 < B_0 < \bar{B}$, is open in the original initial stocks. It is nonempty. In the positive-labor-share cases, choose k_0, d_0 close to $\bar{k}, \bar{d}$ and $\tau_0 > 0$ sufficiently small. Then $A_0 N_0 = 1/\tau_0$, $K_0 = k_0/\tau_0$, and $B_0 = \bar{B} - d_0 \tau_0$ are admissible. In the AI-dominated case, choose d_0 near $\bar{d}$ and $h_0, \tau_0 > 0$ sufficiently small. The inverse relations are $A_0 N_0 = 1/\tau_0$, $K_0 = 1/(\tau_0 h_0^{1/\varphi})$, and $B_0 = \bar{B} - d_0 \tau_0 h_0^{1/\varphi}$. Any positive factorization of $A_0 N_0$ supplies A_0, N_0 . These choices also satisfy the strict efficiency restriction in Step 4.

For each initial state in the resulting set, the local graph selects $C_0, q_0 > 0$. These jump values are unique among the trajectories that remain near the indicated normalized limit and converge to it. The argument does not assert global uniqueness.

For $\sigma < 1$ with $\bar{B} > \mathcal{B}(\sigma)$, and for $\sigma > 1$ with $\bar{B} < \mathcal{B}(\sigma)$, Step 1 gives a unique interior $\bar{s}_X$. Step 2 then yields $g_Y - n \rightarrow \gamma$, $g_w \rightarrow \gamma$, and $r \rightarrow \rho + \gamma$. At $\sigma = 1$, the exact CES limit gives $s_X = \omega_X$ at every date, and the same construction works for every finite positive upper bound. These establish the first three cases of the proposition. Step 3 gives $s_X \rightarrow 1$, $r \rightarrow \mathcal{R}(\bar{B})$, $g_Y - n \rightarrow \mathcal{R}(\bar{B}) - \rho$, and the wage-growth limit in (41), establishing the fourth case. Step 4 verifies the agent-optimality and transversality requirements of Definition 1 for all these trajectories. $\square$

Proof of Proposition 2. Fix $B > 0$ and $\sigma > 1$. Combining marginal-cost pricing, $p_X = 1/B$, with the final-good firm's demand in equation (13) gives

$$\frac{X}{Y} = B(1 - \alpha)s_X \longrightarrow B(1 - \alpha) \quad \text{as } s_X \rightarrow 1. \quad (93)$$

The CES limit in equation (33) is unchanged. Substituting that limit and equation (93) into $Y/K = [(Z/X)(X/Y)]^{(1-\alpha)/\alpha}$, from equation (34), yields

$$\frac{Y}{K} \longrightarrow \left[(1 - \alpha)B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha}.$$

Equation (11) then gives $\mathcal{R}^{\text{mc}}(B)$ in equation (42). Since $0 < 1 - \alpha < 1$, this return strictly exceeds the monopoly return in equation (35).

Both returns are strictly increasing in B , and their formulas imply $\mathcal{R}^{\text{mc}}(B) = \mathcal{R}(B/(1 - \alpha))$. Thus equation (36) gives

$$\mathcal{R}^{\text{mc}}((1 - \alpha)\mathcal{B}(\sigma)) = \mathcal{R}(\mathcal{B}(\sigma)) = \rho + \gamma.$$

Uniqueness of the threshold establishes equation (43). $\square$

Proof of Proposition 3. Equations (45) and (46) follow because $\sigma/(\sigma - 1) < 0$. Define $k = K/(AL)$. The resource constraint (29a), nonnegative consumption and compute expenditure, and $L = N$ imply

$$\dot{k} \leq \omega_L^{\sigma(1-\alpha)/(\sigma-1)} k^\alpha - (\delta + n + \gamma)k. \quad (94)$$

Since $\delta + n + \gamma > 0$ and $0 < \alpha < 1$, the right-hand side is negative above a finite positive level. In particular,

$$k_t \leq \max \left\{ k_0, \left[\frac{\omega_L^{\sigma(1-\alpha)/(\sigma-1)}}{\delta + n + \gamma} \right]^{1/(1-\alpha)} \right\}.$$

Equation (46) then bounds $Y/(AL)$ uniformly. Since $A_t = A_0 e^{\gamma t}$ and $L = N$, Y/N is bounded above by a constant times $e^{\gamma t}$. Taking logarithms, dividing by t , and taking the upper limit gives (47). $\square$

Proof of Proposition 4. All auxiliary variables below are confined to this proof. Write $x = X/(AL)$, $k = K/(AL)$, $c = C/(AL)$, $y = Y/(AL)$, $u = U/(AL)$, $m = M/(AL)$, $d = \rho - n > 0$, and $\nu = \delta + n + \gamma$. No convergence of a growth rate or allocation ratio is assumed.

Step 1. Static optimality gives a limiting production function. Put $\varphi = (\sigma - 1)/\sigma < 0$ and $f(x) = [\omega_L + \omega_X x^\varphi]^{(1-\alpha)/\varphi}$, so $y = k^\alpha f(x)$. The monopoly condition (19) requires $e_X = (1 - s_X)/\sigma + \alpha s_X < 1$. Thus

$$s_X > s_\infty := \frac{1 - \sigma}{1 - \alpha\sigma}, \quad 0 < x < x_\infty := \left[\frac{\omega_L s_\infty}{\omega_X (1 - s_\infty)} \right]^{1/\varphi}. \quad (95)$$

Define $f_\infty = f(x_\infty) > 0$. In particular, $y \leq f_\infty k^\alpha$ and $0 < u \leq x_\infty/B$.

The monopoly condition can be written

$$k^\alpha v(x) = 1/B, \quad v(x) = \frac{(1 - \alpha)s_X f(x)(1 - e_X)}{x}.$$

On $(0, x_\infty)$, v is strictly decreasing, from infinity to zero. Indeed, with a prime here denoting differentiation with respect to $\log x$, $e'_X = (\alpha - \sigma^{-1})\varphi s_X(1 - s_X) > 0$ and $-d \log v / d \log x = e_X + e'_X / (1 - e_X) > \alpha s_X$. Consequently x increases with B and k , while

$$\frac{\partial \log y}{\partial \log k} = \alpha + \frac{\alpha(1 - \alpha)s_X}{e_X + e'_X / (1 - e_X)} < 1.$$

Thus y/k decreases with k and increases with B . For k in any compact interval bounded away from zero, as $B \rightarrow \infty$,

$$x \rightarrow x_\infty, \quad y \rightarrow f_\infty k^\alpha, \quad u \rightarrow 0$$

uniformly. These are properties of the monopoly allocation at given k, B , not assumptions about the equilibrium path. They also imply that, for any prescribed finite return, sufficiently small k and sufficiently large B make r exceed that return, uniformly over those small values of k .

Step 2. Household and developer optimality bound consumption. Proposition 3 bounds k and y above by finite constants $k_{\max}, y_{\max}$. The household Euler equation and $r \geq -\delta$ give $\dot{c}/c \geq -(\delta + \rho + \gamma)$. Integrating the resource constraint over $[t, t+1]$ therefore gives

$$c_t \int_0^1 e^{-(\delta+\rho+\gamma)v} dv \leq \int_t^{t+1} c_s ds \leq k_{\max} + y_{\max}.$$

Hence c has a finite uniform upper bound $c_{\max}$.

Write $D_{t,s} = \exp(-\int_t^s r_v dv)$. Euler implies $\int_t^\infty D_{t,s} C_s ds = C_t/d$. Integrating the household budget (4) and using its TVC (6) shows that the discounted integral of labor income plus developer net profit converges. Since labor income is nonnegative, the discounted integral of developer net profit has a limit that is either finite or minus infinity. The latter is impossible: the developer can stop research and obtain nonnegative operating profits. By continuation optimality, the developer's continuation value $V_t = \int_t^\infty D_{t,s} \Pi_s ds$ is therefore finite and nonnegative. It follows that

$$\frac{C_t}{d} = K_t + V_t + \int_t^\infty D_{t,s} w_s L_s ds, \quad c_t \geq dk_t, \quad \int_t^\infty D_{t,s} w_s L_s ds \leq \frac{C_t}{d}. \quad (96)$$

Here V_t is an auxiliary continuation value, not an additional traded asset or a change to the household budget.

Step 3. Discounted research spending vanishes relative to effective labor. Fix a positive rate a . Step 1 permits a small $k_a > 0$ and a sufficiently late date such that $r - (n + \gamma) \geq a$ whenever $k \leq k_a$. On $[k_a, k_{\max}]$, the wage equation (12) and the static monopoly allocation give a uniform positive lower bound ℓ_a on w/A . Since $r \geq -\delta$, there is a finite constant $H_a > 0$ such that

$$a \leq r - (n + \gamma) + H_a w/A$$

at every sufficiently late date. Multiplication by $D_{t,s} A_s L_s$, integration, and (96) give

$$\int_t^\infty D_{t,s} A_s L_s ds \leq \frac{A_t L_t + H_a C_t/d}{a} \leq J A_t L_t \quad (97)$$

for a fixed finite J . The omitted terminal contribution from discounted effective labor is nonpositive; no discount-rate limit is used.

The research condition (22) and costate equation (23) imply

$$\frac{d(qB)}{dt} = r q B - U + \frac{1-\eta}{\eta} M. \quad (98)$$

Since B is nondecreasing, $U_s \leq x_\infty A_s L_s / B_t$ for $s \geq t$. Integrating (98) first over finite intervals and then using the developer TVC (25) gives

$$q_t B_t + \frac{1-\eta}{\eta} \int_t^\infty D_{t,s} M_s ds = \int_t^\infty D_{t,s} U_s ds \leq \frac{x_\infty J}{B_t} A_t L_t. \quad (99)$$

The right side is finite by (97); nonnegativity then ensures that the research integral is finite as well. Its value divided by $A_t L_t$ tends to zero. This argument requires $0 < \eta < 1$, but neither $\eta \leq 1/2$ nor $\eta < \alpha$.

Step 4. Capital cannot repeatedly approach zero. Let $F(k) = f_\infty k^\alpha - \nu k$. Choose $\varepsilon > 0$ small enough that F is positive and increasing on $(0, \varepsilon]$. Step 1 also permits a sufficiently late date and $a > 0$ such that $\dot{c}/c = r - \rho - \gamma \geq a$ whenever $k \leq \varepsilon$. For any $j \leq \varepsilon$, the equilibrium path cannot enter $\{k \leq j, c \geq F(j)\}$: there $\dot{k} \leq F(k) - c \leq 0$, whereas c continues to grow at least at rate a , contradicting its upper bound.

If k stayed below ε forever, consumption would likewise exceed its upper bound. Hence the path eventually reaches ε . Suppose arbitrarily late crossings from ε to $\varepsilon/2$ were possible. Let t_0 be the last crossing of ε before the first subsequent crossing t_1 of $\varepsilon/2$. Throughout this interval $k \in [\varepsilon/2, \varepsilon]$ and c increases. Equation (96) gives $c_{t_0} \geq d\varepsilon$, so

$$t_1 - t_0 \leq T_\varepsilon := a^{-1} \log\left(\frac{c_{\max}}{d\varepsilon}\right) < \infty.$$

At t_1 , the preceding invariant-region argument requires $c_{t_1} < F(\varepsilon/2)$. Thus $c_s < F(\varepsilon/2)$ throughout the crossing. The uniform convergence in Step 1, applied on the crossing interval, yields numbers $\epsilon_{t_0} \rightarrow 0$ such that

$$\dot{k}_s = y_s - u_s - \nu k_s - c_s - m_s \geq -\epsilon_{t_0} - m_s.$$

On this interval r is bounded above by a fixed positive R_ε . Since $n + \gamma > 0$, equation (99) then implies

$$\frac{\varepsilon}{2} \leq T_\varepsilon \epsilon_{t_0} + \int_{t_0}^{t_1} m_s ds \leq T_\varepsilon \epsilon_{t_0} + e^{R_\varepsilon T_\varepsilon} \frac{\eta x_\infty J}{(1 - \eta) B_{t_0}} \rightarrow 0,$$

a contradiction. Therefore k is eventually bounded away from zero, and so is $c \geq dk$. In particular, r is now bounded above.

Step 5. Research per unit of effective labor vanishes and Ramsey dynamics determine the limit. Differentiating the research condition and using the costate equation gives

$$(1 - \eta)g_M = r - \frac{\eta \chi^U B^{\eta-1}}{M^{1-\eta}} \leq r. \quad (100)$$

Choose a positive R above all eventual values of r and put $H = R/(1 - \eta)$. Then $M_s \geq M_t e^{-H(t-s)}$ on $[t-1, t]$ at sufficiently late dates. Consequently

$$\frac{1}{A_{t-1} L_{t-1}} \int_{t-1}^t D_{t-1,s} M_s ds \geq e^{n+\gamma-R-H} m_t.$$

The left side tends to zero by (99). Hence $m \rightarrow 0$, and Step 1 already gives $u \rightarrow 0$.

The normalized equilibrium equations therefore approach, uniformly on the compact positive

region containing the path, the Ramsey system

$$\dot{k} = F(k) - c, \quad \dot{c} = c[\alpha f_\infty k^{\alpha-1} - \delta - \rho - \gamma]. \quad (101)$$

For clarity, the following argument verifies convergence rather than inferring it solely from convergence of the equations. This limiting system has one positive stationary point,

$$k_\infty = \left[\frac{\alpha f_\infty}{\rho + \gamma + \delta} \right]^{1/(1-\alpha)}, \quad c_\infty = F(k_\infty) > 0.$$

Its Jacobian has negative determinant, so it is a saddle. The standard Ramsey phase portrait is discussed by [Groth \(2013\)](#). There is no nonconstant complete orbit confined to a compact subset of the positive quadrant. To see this, the multiplier $1/c^2$ gives

$$\frac{\partial}{\partial k} \frac{F(k) - c}{c^2} + \frac{\partial}{\partial c} \frac{\alpha f_\infty k^{\alpha-1} - \delta - \rho - \gamma}{c} = \frac{\rho - n}{c^2} > 0.$$

The Bendixson–Dulac argument excludes both periodic orbits and homoclinic loops. The planar limit-set theorem, the unique stationary point, and its saddle character then exclude a nonconstant complete orbit confined to such a compact set: its forward and backward limits would otherwise require a periodic orbit or a connection back to that same saddle.

For any sequence of dates tending to infinity, time translations of our bounded positive path have a subsequence converging on every bounded time interval to a complete orbit of (101). This follows from bounded derivatives, uniform convergence of the vector fields, and the integral form of the differential equations. The preceding paragraph forces that orbit to be the stationary point. Thus every accumulation point is (k_∞, c_∞) , proving convergence of the original path.

It follows that $y \rightarrow y_\infty = f_\infty k_\infty^\alpha > 0$ and

$$r \rightarrow \rho + \gamma, \quad \frac{K}{Y} \rightarrow \frac{\alpha}{\rho + \gamma + \delta}, \quad \frac{C}{Y} \rightarrow 1 - \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} > 0. \quad (102)$$

Positivity follows because the numerator of the final expression is $\rho - n + (1 - \alpha)(\delta + n + \gamma) > 0$. This proves (48). Since $u, m \rightarrow 0$ and $y \rightarrow y_\infty > 0$, it follows that $U/Y = u/y \rightarrow 0$ and $M/Y = m/y \rightarrow 0$, proving (49). It also gives $\dot{k} \rightarrow 0$ and $g_C \rightarrow n + \gamma$.

Step 6. Output and wage growth converge as well. A convergent level need not have a convergent derivative. The bound on g_M from Step 5 and the RSI technology give

$$B_t^{1-\eta} \geq (1 - \eta)\chi M_t^\eta \int_0^1 e^{-\eta H v} dv.$$

Thus $g_B = \chi M^\eta / B^{1-\eta}$ is bounded above. At x_∞ , the marginal-revenue function v defined in Step 1 has a simple zero: $v(x_\infty) = 0$ and $v'(x_\infty) < 0$, where the prime now denotes differentiation with

respect to x . Differentiating $k^\alpha v(x) = 1/B$ yields

$$\dot{x} = -\frac{v(x)}{v'(x)} \left(g_B + \alpha \frac{\dot{k}}{k} \right) \rightarrow 0.$$

Hence $\dot{y} \rightarrow 0$ and $\dot{s}_X \rightarrow 0$. Since $y_\infty > 0$ and $1 - s_\infty > 0$, $g_{Y/N} \rightarrow \gamma$ and, by (12), $g_w \rightarrow \gamma$. The limiting income shares follow by substituting s_∞ into (12) and (13). The result is conditional on equilibrium existence and $B \rightarrow \infty$, not an equilibrium-existence theorem or an assumption of balanced growth. $\square$

Proof of Proposition 5. Use the definitions in Appendix C. *The reference BGP.* First, all candidate ratios are positive. The denominator of the research share m^* , $\rho - n + \eta g_Y^*$, is positive. Moreover,

$$i^* = \frac{\alpha(g_Y^* + \delta)}{\rho - n + g_Y^* + \delta} < \alpha, \quad m^* < \frac{u^* \eta}{1 - \eta}.$$

Since $\Delta = (1 - \alpha)(1 - \eta - \omega_X) > 0$, $\omega_X < 1 - \eta$ and $\omega_X < 1$. Consequently,

$$u^* + m^* < \frac{u^*}{1 - \eta} < (1 - \alpha)\omega_X < 1 - \alpha, \quad c^* = 1 - u^* - i^* - m^* > 0.$$

Thus positive consumption follows from the stated conditions; it is not an additional assumption.

For given $A_0, N_0 > 0$, choose the reference initial stocks as

$$B_0^* = \left[A_0 N_0 (k^*)^{\alpha/[(1-\alpha)\omega_L]} (u^*)^{\omega_X/\omega_L} m^* \left(\frac{\chi}{g_B^*} \right)^{1/\eta} \right]^{\eta\omega_L/(1-\eta-\omega_X)}, \quad (103)$$

$$Y_0^* = A_0 N_0 (k^*)^{\alpha/[(1-\alpha)\omega_L]} (u^* B_0^*)^{\omega_X/\omega_L}, \quad K_0^* = k^* Y_0^*. \quad (104)$$

The exponent is positive. These choices solve simultaneously the initial production equation and $g_B^* B_0^* = \chi(B_0^* m^* Y_0^*)^\eta$. Set $U_0^* = u^* Y_0^*$, $M_0^* = m^* Y_0^*$, $C_0^* = c^* Y_0^*$, $X_0^* = B_0^* U_0^*$, and

$$q_0^* = \frac{M_0^*}{\eta g_B^* B_0^*}. \quad (105)$$

Let $Y_t^* = Y_0^* e^{g_Y^* t}$, $B_t^* = B_0^* e^{g_B^* t}$, $K_t^* = k^* Y_t^*$, $C_t^* = c^* Y_t^*$, $U_t^* = u^* Y_t^*$, $M_t^* = m^* Y_t^*$, and $q_t^* = q_0^* e^{(g_Y^* - g_B^*) t}$. Use the exogenous A_t, N_t and set $L_t = N_t$ and $X_t^* = B_t^* U_t^*$. The static equilibrium equations determine all remaining quantities and prices.

The two growth identities (54) and (55) ensure that the production equation and the RSI technology hold at every date once they hold initially. The static monopoly condition gives $p_X X = (1 - \alpha)\omega_X Y$ and $U = u^* Y$. Equation (105) gives $M = \eta q \dot{B}$, which is the research first-order condition (28k) with $\psi = 1$. The definitions of k^* , r^* , and c^* verify the capital-return, Euler, and

resource equations. The costate equation reduces to

$$g_Y^* - g_B^* = r^* - \frac{u^* \eta g_B^*}{m^*} - \eta g_B^*.$$

This holds by equation (134) and $(1 - \eta)g_B^* = \eta g_Y^*$. All allocations are positive and finite at every finite date.

Both transversality conditions are explicit:

$$e^{-\rho t} \frac{N_t K_t^*}{C_t^*} = \frac{N_0 k^*}{c^*} e^{-(\rho-n)t} \longrightarrow 0, \quad (106)$$

$$e^{-r^* t} q_t^* B_t^* = q_0^* B_0^* e^{-(\rho-n)t} \longrightarrow 0. \quad (107)$$

Discounted developer profit is a constant multiple of $e^{-(\rho-n)t}$. Discounted household utility is $N_0 e^{-(\rho-n)t} [\log(C_0^*/N_0) + (g_Y^* - n)t]$. Both improper objective integrals are finite.

It remains to establish global optimality, rather than only satisfaction of the necessary conditions. At given paths of K, A, L , static optimization gives

$$\begin{aligned} \pi_t(B) &= (1 - \alpha)\omega_X [1 - (1 - \alpha)\omega_X] [K_t^\alpha (A_t L_t)^{(1-\alpha)\omega_L}]^{1/[1-(1-\alpha)\omega_X]} \\ &\quad \times [(1 - \alpha)^2 \omega_X^2 B]^{(1-\alpha)\omega_X/[1-(1-\alpha)\omega_X]}. \end{aligned} \quad (108)$$

To verify developer optimality by concavity, define the invertible state transformation and its current-value costate by

$$\nu = \frac{1 - \eta}{\eta}, \quad S = B^\nu, \quad p = \frac{q}{\nu B^{\nu-1}}. \quad (109)$$

The RSI technology becomes

$$\dot{S} = \nu \chi S^{1-\eta} M^\eta,$$

which is jointly concave in S, M for every $0 < \eta < 1$. At given K, A, L , equation (108) is a positive time-dependent coefficient times

$$S^{\frac{(1-\alpha)\omega_X}{\nu[1-(1-\alpha)\omega_X]}}.$$

Since $\eta + \omega_X < 1$ implies $(1 - \alpha)\omega_X < 1 - \eta$, this exponent lies strictly between zero and one. Operating profit is therefore concave in S . With $p > 0$, the transformed Hamiltonian $\pi_t(S^{1/\nu}) - M + p_t \nu \chi S^{1-\eta} M^\eta$ is jointly concave. The research condition (22) and costate equation (23), with $\psi = 1$, give its control first-order condition and costate equation by the chain rule.

Write $D_t = \exp(-\int_0^t r_s ds)$ and $J_T = \int_0^T D_t [\pi_t(B_t) - M_t] dt$. For any admissible alternative with the same initial B , concavity, the research condition, and the costate equation give, after integration by parts,

$$\tilde{J}_T - J_T \leq -D_T p_T (\tilde{S}_T - S_T) \leq D_T p_T S_T = \frac{D_T q_T B_T}{\nu} \longrightarrow 0. \quad (110)$$

The last limit follows from the developer TVC. Since the transformation preserves feasible research paths and their payoffs, no admissible alternative has a greater limiting payoff, or a greater upper

limit of truncated payoff, than this finite-valued candidate. The same verification applies to nearby candidates satisfying the uncapped equations and the developer TVC; it does not require exact balanced growth.

For the household, set $\Lambda_t = e^{-\rho t} N_t / C_t$. Euler implies $\dot{\Lambda} = -r\Lambda$. Concavity of log utility and the linear budget give, for any feasible alternative with the same K_0 and the same given prices and developer net profit,

$$\int_0^T e^{-\rho t} N_t \left[\log(\tilde{C}_t / N_t) - \log(C_t / N_t) \right] dt \leq -\Lambda_T (\tilde{K}_T - K_T) \leq \Lambda_T K_T \rightarrow 0.$$

This establishes global household optimality. Competitive final-good firms face a concave constant-returns technology, and Lemma 1 establishes global static monopoly optimality. The constructed path satisfies Definition 1 with the uncapped RSI technology (44).

Local convergence from nearby initial stocks. Use the ordering $\xi = (\xi_K, \xi_B, \xi_C, \xi_q)'$ in (136), and let e_K, e_B, e_C, e_q be the corresponding unit row vectors. Define the row vectors

$$y_\xi = \left(\frac{\alpha}{1 - (1 - \alpha)\omega_X}, \frac{(1 - \alpha)\omega_X}{1 - (1 - \alpha)\omega_X}, 0, 0 \right), \quad (111)$$

$$m_\xi = \frac{\eta e_B + e_q}{1 - \eta}, \quad (112)$$

$$b_\xi = g_B^*[(\eta - 1)e_B + \eta m_\xi]. \quad (113)$$

The reduced production equation gives the log deviation of output as $y_\xi \xi$. The research FOC implies $M = (q\chi\eta B^\eta)^{1/(1-\eta)}$, so the log deviation of M is $m_\xi \xi$. Differentiating $g_B = \chi B^{\eta-1} M^\eta$ at the BGP gives the level deviation $b_\xi \xi$.

The four rows of J^* , derived from capital accumulation, AI research, the household Euler equation, and the developer costate equation, are

$$J_K^* = \frac{1 - u^*}{k^*} y_\xi - (g_Y^* + \delta) e_K - \frac{c^*}{k^*} e_C - \frac{m^*}{k^*} m_\xi, \quad (114)$$

$$J_B^* = b_\xi, \quad (115)$$

$$J_C^* = (r^* + \delta)(y_\xi - e_K), \quad (116)$$

$$J_q^* = (r^* + \delta)(y_\xi - e_K) - (\rho - n + \eta g_Y^*)(y_\xi - e_q - e_B) - \eta b_\xi. \quad (117)$$

The last row uses $U/(qB) = \rho - n + \eta g_Y^*$ at the BGP. All entries depend on growth rates and ratios, which are independent of χ . No terminal restrictions were used to derive this Jacobian.

I next establish the two local properties (137) and (138) from the parameter restrictions, rather than impose them. Retain ν from (109) and, for this linearization, define

$$\begin{aligned} a_K &= \frac{\alpha}{1 - (1 - \alpha)\omega_X}, & a_T &= \frac{(1 - \alpha)\omega_X \eta}{[1 - (1 - \alpha)\omega_X](1 - \eta)}, \\ d &= \rho - n, & P &= r^* + \delta, & h &= \rho - n + \eta g_Y^*. \end{aligned}$$

All six quantities are positive, and the restriction $\eta + \omega_X < 1$ gives

$$1 - a_K - a_T = \frac{(1 - \alpha)(1 - \eta - \omega_X)}{[1 - (1 - \alpha)\omega_X](1 - \eta)} > 0. \quad (118)$$

Make the invertible linear change of deviations

$$\widehat{\xi} = (\xi_K, \nu\xi_B, \xi_C, \xi_q + (1 - \nu)\xi_B)'$$

The second and fourth coordinates are the log deviations of the transformed state S and its shadow price p . This changes neither the model nor the eigenvalues. Applying it to (114)–(117) gives a similar matrix

$$\widehat{J} = \begin{pmatrix} \mathbf{A} & \mathbf{B} & -\mathbf{C} & -\mathbf{V} \\ 0 & 0 & 0 & \mathbf{D} \\ -\mathbf{G} & \mathbf{H} & 0 & 0 \\ -\mathbf{E} & \mathbf{F} & 0 & d \end{pmatrix}, \quad (119)$$

where the following letters abbreviate matrix coefficients only:

$$\begin{aligned} \mathbf{A} &= d + (1 - \alpha)\omega_X P, & \mathbf{B} &= \frac{(1 - u^*)a_T - m^*}{k^*}, & \mathbf{C} &= \frac{c^*}{k^*}, \\ \mathbf{V} &= \frac{m^*}{k^*(1 - \eta)}, & \mathbf{D} &= \frac{\eta g_Y^*}{1 - \eta}, & \mathbf{G} &= P(1 - a_K), \\ \mathbf{H} &= P a_T, & \mathbf{E} &= \mathbf{G} + h a_K, & \mathbf{F} &= \mathbf{H} + h(1 - a_T). \end{aligned}$$

All coefficients other than $\mathbf{B}$ are positive by their definitions. The BGP allocation ratios imply the identities

$$\mathbf{L} \equiv \mathbf{B} - \frac{\mathbf{V}d}{\mathbf{D}} = \frac{(1 - \alpha)\omega_X \eta}{(1 - \eta)k^*} > 0, \quad (120)$$

$$\mathbf{G}\mathbf{F} - \mathbf{H}\mathbf{E} = hP(1 - a_K - a_T) > 0. \quad (121)$$

Set $\mathbf{N} = \mathbf{C}\mathbf{G} + \mathbf{V}\mathbf{E} > 0$ and $\mathbf{Q} = \mathbf{C}\mathbf{H} + \mathbf{V}\mathbf{F} > 0$. Expanding the determinant gives the characteristic polynomial

$$\begin{aligned} p(\ell) &= \det(\ell I - \widehat{J}) \\ &= (\ell^2 - \mathbf{A}\ell - \mathbf{N})(\ell^2 - d\ell - \mathbf{D}\mathbf{F}) - \mathbf{D}\mathbf{E}(\mathbf{Q} - \mathbf{L}\ell). \end{aligned} \quad (122)$$

In particular,

$$p(0) = \mathbf{D}\mathbf{C}hP(1 - a_K - a_T) > 0.$$

Let $\mu = (d - \sqrt{d^2 + 4\mathbf{D}\mathbf{F}})/2 < 0$, the negative root of the second quadratic factor. Then $p(\mu) = -\mathbf{D}\mathbf{E}(\mathbf{Q} - \mathbf{L}\mu) < 0$. Since $p(\ell) \rightarrow +\infty$ as $\ell \rightarrow -\infty$, there is one negative root below μ and another between μ and zero. These are the only negative roots: the cubic coefficient of p is $-(\mathbf{A} + d) < 0$, its linear coefficient is $\mathbf{A}\mathbf{D}\mathbf{F} + d\mathbf{N} + \mathbf{L}\mathbf{D}\mathbf{E} > 0$, and its constant coefficient is positive. Thus the

coefficient signs of $p(-x)$, in descending powers, are $(+, +, \text{arbitrary}, -, +)$. Descartes' rule of signs gives exactly two sign changes, regardless of the quadratic coefficient. The two negative roots are therefore real, distinct, and simple. Write them as $\ell_1 < \mu < \ell_2 < 0$. The remaining two roots have positive product and positive sum, because $p(0) > 0$ and the sum of all four roots is $A + d > 0$. If real, both are positive; if complex, their real parts are positive. This proves (137).

To establish nonsingularity of the projection onto the two stocks, write a stable eigenvector in transformed coordinates as $(x, y, z, u)'$. Its second component cannot be zero: the second, fourth, and first rows would successively imply $u = x = z = 0$. Normalize $y = 1$. The second and fourth rows then give

$$x(\ell) = \frac{\text{DF} + d\ell - \ell^2}{\text{DE}}.$$

The ordering of the roots implies $x(\ell_1) < 0 < x(\ell_2)$. Hence the two projected stable eigenvectors $(x(\ell_1), 1)'$ and $(x(\ell_2), 1)'$ are linearly independent. Returning to (ξ_K, ξ_B) only rescales their second component by $1/\nu > 0$, so (138) follows as well.

The exact normalized vector field is continuously differentiable near $\xi = 0$. The spectral property just proved and the stable-manifold theorem supply a two-dimensional local manifold whose trajectories remain near zero and converge exponentially. The derivative of its projection onto (ξ_K, ξ_B) is V_{KB} . The nonsingularity just proved and the inverse-function theorem therefore make this manifold locally the graph of a continuously differentiable function selecting (ξ_C, ξ_q) from the two initial stock deviations. Since exponentiation is a local diffeomorphism, the same holds for (C_0, q_0) as functions of (K_0, B_0) . More explicitly, for some constants $a, H > 0$, trajectories on a sufficiently small stable manifold satisfy

$$\|\xi(t)\| \leq H e^{-at} \|\xi(0)\|, \quad t \geq 0.$$

For any sufficiently small open neighborhood $\mathcal{V}$ of zero, choose a ball contained in $\mathcal{V}$. Continuity of the graph selecting the two jump variables permits an open neighborhood $\mathcal{U}$ of the initial stock deviations for which $H\|\xi(0)\|$ is smaller than that ball's radius. Every initial stock pair in $\mathcal{U}$ therefore has a trajectory staying in $\mathcal{V}$ for all $t \geq 0$ and converging to zero. Exponentiation maps $\mathcal{U}$ to an open neighborhood of (K_0^*, B_0^*) . Shrinking these neighborhoods preserves positivity, feasibility, and existence for every $t \geq 0$. Local uniqueness follows because every trajectory remaining in this local region and converging to zero belongs to the same stable manifold, which is a graph over stocks. This does not assert convergence from arbitrary initial C_0, q_0 or exclude equilibrium trajectories that leave the region.

The statement also permits nearby A_0, N_0 . Equations (103) and (104) make the reference B_0^*, K_0^* smooth functions of these two positive initial values. After normalization by the corresponding BGP, the exact vector field depends only on the parameters and BGP ratios, not on A_0, N_0 . The set of (A_0, N_0, K_0, B_0) whose stock deviations belong to $\mathcal{U}$ is therefore open and contains the reference initial condition. From each such initial condition, the construction converges to the BGP associated with its own A_0, N_0 , not necessarily to the same levels as the original reference path.

Since $f(\xi(t)) \rightarrow 0$, growth rates converge to their BGP values, not only normalized levels. The static equations give the same convergence for the interest rate, wage growth, and allocation ratios. The unit-elastic labor income share remains $(1 - \alpha)\omega_L$ at every date.

Exponential convergence implies that $r_t - r^*$ is integrable. Thus D_t/e^{-r^*t} tends to a finite positive constant, and both TVC expressions equal their BGP counterparts in (106) and (107) times factors tending to finite positive limits. Both TVCs hold. Output and all expenditure flows are bounded multiples of $e^{g_Y^*t}$, while log consumption per person is a linear function of time plus a bounded deviation. Both objective integrals therefore remain finite. The global concavity verification established above applies at these new price and allocation paths, proving agent optimality. Static optimality and market clearing then establish equilibrium, not merely a solution of the necessary conditions. $\square$

Proof of Proposition 6. I construct a sequence of research trajectories whose discounted net profit grows without bound, although each trajectory has finite expenditure and AI efficiency. Write $\theta = (1 - \alpha)/\alpha$. For a research trajectory $\{M_t\}_{t \in [0, T]}$, discounted net profit on the fixed horizon is

$$J_T(M) = \int_0^T \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt.$$

Integrability of r bounds the discount factor above and away from zero on $[0, T]$. At the optimal service choice, equations (13) and (19) imply

$$\pi(B) = p_X X - \frac{X}{B} = e_X p_X X = (1 - \alpha) e_X s_X Y.$$

For $\sigma > 1$, marginal revenue is positive at every finite X and approaches zero as $X \rightarrow \infty$, as established in Lemma 1. The monopoly condition therefore implies $X \rightarrow \infty$ as $B \rightarrow \infty$, holding K, A, L fixed. Consequently, $s_X \rightarrow 1$ and $e_X \rightarrow \alpha$. Substituting these limits into equation (62) gives

$$\pi(B) \sim \kappa_X K B^\theta, \quad \kappa_X = \alpha(1 - \alpha) \left[(1 - \alpha)^2 \omega_X^{\sigma/(\sigma-1)} \right]^\theta > 0.$$

The operating-profit asymptotic is uniform over dates in $[0, T]$ because K, A, L are continuous, bounded, and bounded away from zero on $[0, T]$.

Fix $\Delta \in (0, T)$, choose $M = \mathcal{M}$ on $[0, \Delta]$, and choose $M = 0$ afterward. The uncapped RSI technology gives

$$B_{\mathcal{M}}(t)^{1-\eta} = B_0^{1-\eta} + (1 - \eta)\chi\mathcal{M}^\eta \min\{t, \Delta\}.$$

Since optimized operating profit is nonnegative, a lower bound on discounted net profit counts only operating profits earned after research stops, while subtracting all research expenditure. For all sufficiently large $\mathcal{M}$,

$$J_T(\mathcal{M}) \geq c_\pi \mathcal{M}^{\eta(1-\alpha)/[\alpha(1-\eta)]} - c_M \mathcal{M}, \quad (123)$$

where $c_\pi, c_M > 0$ are independent of $\mathcal{M}$. The exponent exceeds one because $\eta > \alpha$, so $J_T(\mathcal{M}) \rightarrow +\infty$

as $\mathcal{M} \rightarrow \infty$. Each finite $\mathcal{M}$ keeps AI efficiency and discounted net profit finite on $[0, T]$; it is the sequence of payoffs that is unbounded. If research expenditure follows a candidate equilibrium's trajectory after T , subsequent operating profits cannot be lower once the initial research phase leaves AI efficiency above the candidate's level at T . $\square$

Proposition 7 (An AI bottleneck under complementarity). *Suppose $0 < \sigma < 1$ and $0 < B_0 < \bar{B} < \mathcal{B}(\sigma)$. Any equilibrium satisfies*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \mathcal{R}(\bar{B}) - \rho < \gamma,$$

$$\frac{X}{AL} \rightarrow 0, \quad s_X \rightarrow 1, \quad \frac{wL}{Y} \rightarrow 0.$$

Proof of Proposition 7. Consider any equilibrium with $0 < \sigma < 1$ and $0 < B_0 < \bar{B} < \mathcal{B}(\sigma)$. The increasing share map in equation (79), established in Step 1 of the proof of Proposition 1, and $B_t \leq \bar{B}$ imply

$$r_t \leq \mathcal{R}(B_t) \leq \mathcal{R}(\bar{B}) < \rho + \gamma, \quad \frac{Y_t}{K_t} \leq \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha}. \quad (124)$$

The present value of the developer's net profit from any date onward is nonnegative: it can stop research and earn nonnegative operating profit. Wage income is also nonnegative. Integrating the household budget (4), using its transversality condition (6) and the Euler equation (5), therefore gives the first inequality below; integrating the Euler equation with the return bound gives the second:

$$C_t \geq (\rho - n)K_t, \quad \frac{C_t}{N_t} \leq \frac{C_0}{N_0} e^{[\mathcal{R}(\bar{B}) - \rho]t}. \quad (125)$$

Combining these inequalities yields

$$\frac{Y_t}{N_t} \leq \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha(\rho - n)} \frac{C_0}{N_0} e^{[\mathcal{R}(\bar{B}) - \rho]t}, \quad \limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \mathcal{R}(\bar{B}) - \rho < \gamma. \quad (126)$$

Consequently $Y/(AL) \rightarrow 0$. Since equation (32) gives $X/Y \leq \bar{B}(1 - \alpha)^2$ under complementarity, $X/(AL) \rightarrow 0$: AI production services become scarce relative to effective labor. Equation (10) then implies $s_X \rightarrow 1$ and hence $wL/Y \rightarrow 0$. The growth bound concerns the long-run average, not convergence of the instantaneous growth rate. Neither convergence of C/Y nor $B_t \rightarrow \bar{B}$ is assumed. These are necessary properties of any equilibrium in this region, not a proof of equilibrium existence. $\square$

B Numerical algorithm and equilibrium audit

B.1 Economic intuition and initial conditions

For the comparisons in Section 6, I compute numerical approximations to equilibrium trajectories by solving a boundary-value problem (BVP). Initial capital K_0 and AI efficiency B_0 are given, whereas

initial consumption C_0 and the developer's shadow price q_0 are determined jointly with the transition. I solve the equilibrium differential equations over a finite horizon, using the given stocks as initial conditions and the long-run characterization in Proposition 1 to construct terminal conditions. The checks described below assess feasibility, optimality, and sensitivity to the computational horizon.

For each σ , I first solve the existing-AI economy with efficiency fixed at B_0 and research unavailable. A scalar root search in $\log K_0$ solves Equation (63) using the production and monopoly equations. The resource constraint (26), with $M = 0$ and $\dot{K} = (n + \gamma)K$, gives pre-event consumption. Homogeneity extends these values to a BGP: K, Y, C, U, X grow at $n + \gamma$, wages grow at γ , and B, p_X, r and income shares are constant. Since $r - (n + \gamma) = \rho - n > 0$, pre-event developer profit has finite present value and the household's transversality expression decays.

At unexpected RSI activation, capital and AI efficiency are inherited, as are the exogenous levels of population and labor-augmenting technology. The positive- χ problem selects consumption, research and the shadow value anew. A separate audit checks continuity of production quantities and prices, verifies the pre-event Euler, resource and monopoly conditions, and records the endogenous jumps in consumption and research.

B.2 Boundary-value solution and reported outcomes

I use `scipy.integrate.solve_bvp` from the open-source Python library SciPy (Virtanen et al., 2020). The solver implements fourth-order collocation with residual control similar to that of Kierzenka and Shampine (2001). I use the computational coordinates

$$\left(\log K, \log \frac{B}{\bar{B} - B}, \log C, \log q \right).$$

The second coordinate expresses AI efficiency relative to the remaining gap below its upper bound; it can grow without bound while AI efficiency remains bounded. Subtracting the analytical terminal aggregate growth rate times t from each coordinate removes its asymptotic linear trend. These transformations change the numerical representation, not the economic variables or calendar time. At each trial state the code solves the static monopoly problem and recovers research from its first-order condition. The left boundary fixes K_0, B_0 ; two right-boundary projections remove unstable components of the normalized equilibrium dynamics near the long-run limit, using Appendix A.

For each of the eight parameter configurations, I start near its analytical limiting point and move the initial stocks to their prescribed values in 32 continuation stages. Neither $\omega_X = 0$ nor $\chi = 0$ is used as the starting point for continuation. These stages change the initial conditions of the numerical problem; they are not dates on the economic trajectory. At unit elasticity the static block uses its exact Cobb–Douglas limit; near one, `log1p` and `expm1` avoid cancellation. Under complementarity, marginal revenue is evaluated directly and the static root is polished when needed near zero marginal revenue; the first-order condition is unchanged.

The values $\chi = 7.5$ and $\chi = 1.5$ are supplied directly: there is no outer calibration search. Research shares and price declines are outputs. Annual research expenditure relative to output is

measured as $\int_0^1 M_t dt / \int_0^1 Y_t dt$, not M_0/Y_0 . Gauss–Legendre quadrature with 64 and 128 nodes must agree within 10^{-8} in logs. The annual ratio must also change by less than 2×10^{-5} in logs across the horizon extensions.

B.3 Equilibrium checks

I extend each computational horizon twice by 500 years and tighten the collocation tolerance from 2×10^{-6} to 10^{-8} and 10^{-9} . The final boundary tolerance is 10^{-11} . Initial meshes contain 221, 401 and 601 points and adapt as needed. The displayed window lies inside each solved horizon; no spline is extrapolated.

The original equations are reconstructed independently with five-point differences at 1,001 dates and steps of 0.003 and 0.001 years. An additional 801-date check over the first ten years uses steps of 0.0003 and 0.0001. Acceptance of a computed trajectory requires dynamic residuals below 10^{-6} , first-order residuals below 10^{-9} , final-horizon changes below 2×10^{-5} in detrended logs, and terminal-coordinate gaps below 10^{-4} . The horizon comparison measures changes in the computed trajectory when the terminal date is extended; the terminal-coordinate check measures its distance from the analytical limiting point at that date. These are the existing numerical tolerances, not relaxed economic conditions.

The audit checks feasibility and developer global optimality. When the concavity conditions of Lemma 3 fail, it checks the Hamiltonian support condition in Lemma 2, using counterfactual-state minimization and an analytical bound beyond the grid. At each checked date, the support test compares the maximized Hamiltonian at alternative AI-efficiency levels with its tangent at the candidate state, holding the transformed shadow price and the dated values of capital and effective labor fixed. The two grids contain 81 dates and 101 alternative states, and 321 dates and 241 alternative states, respectively, supplemented by dense first-decade checks when needed. The analytical support bound and the asymptotic rate $n - \rho < 0$ of both transversality expressions support the infinite-horizon continuation. These are numerically verified equilibrium approximations, not interval-arithmetic existence certificates for arbitrary initial stocks.

Table 3: Numerical accuracy of the illustrative RSI comparisons

χ	σ	Final horizon	Mesh points	Dynamic residual	Horizon change
7.5	0.90	5,428.0	1,193	1.18×10^{-9}	3.06×10^{-8}
7.5	1.00	4,944.5	1,290	1.38×10^{-9}	3.46×10^{-8}
7.5	1.10	4,828.8	1,326	1.66×10^{-9}	3.73×10^{-8}
7.5	1.50	3,013.0	1,586	1.69×10^{-9}	1.56×10^{-8}
1.5	0.90	6,047.0	841	1.02×10^{-9}	4.30×10^{-8}
1.5	1.00	5,563.5	893	1.11×10^{-9}	3.80×10^{-8}
1.5	1.10	5,447.8	911	1.19×10^{-9}	4.08×10^{-8}
1.5	1.50	3,632.0	1,104	1.23×10^{-9}	1.80×10^{-8}

Dynamic residuals are the largest independently reconstructed errors across the full-horizon and dense first-decade checks. Horizon change compares the last two solutions in detrended logarithmic coordinates over years 0–500. Error bounds are rounded upward. All eight paths pass the equation, event-continuity, developer-optimality and long-run-continuation checks. The $\sigma = 1.50$ cases use the global Hamiltonian-support test and its analytical continuation bound; the other cases pass the concavity test.

B.4 Analytical checks on developer optimality

At unit elasticity, the operating-profit concavity condition in Lemma 3 has a simple primitive form. Define $\beta = (1 - \alpha)\omega_X$. The static monopoly solution implies $\pi_t(B) = a_t B^{\beta/(1-\beta)}$ for some $a_t > 0$, conditional on the dated values of capital and effective labor. Here R_B is the transformed AI-efficiency state in equation (67). Because $B(R_B)$ is increasing and concave, $R_B \mapsto \pi_t(B(R_B))$ is globally concave whenever $\beta \leq 1/2$. In the benchmark calibration in Table 2, $\beta = 0.067$, so this sufficient condition holds with substantial slack. The research-term condition (72) also holds because $\eta = 0.20 < 1/2$. These are sufficient conditions, not additional maintained assumptions; when they fail, the support condition can instead be checked directly.

The AI-dominated continuation supplies an analytical long-run check on the support inequality. For $1 < \sigma \leq 1/\alpha$, the service elasticity defined in equation (73) satisfies $\varepsilon_{B,t}(b) \leq 1/\alpha$ for every t and b . Suppressing these arguments, subtracting $\alpha(1 - e_X)$ from the denominator in that equation gives

$$(\sigma^{-1} - \alpha)(1 - s_X) [(1 - \sigma^{-1})(1 - s_X) + (\sigma^{-1} - \alpha)s_X] \geq 0.$$

Choose $B_* \in (B_0, \bar{B})$ with $B_*/\bar{B} > \max\{0, (1 - 2\alpha)/(1 - \alpha)\}$. Operating profit is then concave in the transformed state R_B for $B \geq B_*$ by equation (75). If $\eta \leq 1/2$, the research term in (69) is concave there as well. For a candidate with $B_t \geq B_*$, it consequently suffices to check

$$\hat{\mathcal{H}}_t(R_{B,t}) - \hat{\mathcal{H}}'_t(R_{B,t})(R_{B,t} - R_{B,0}) \geq \hat{\mathcal{H}}_t(R_B(B_*)). \quad (127)$$

Above B_* the tangent bound follows from concavity. Below B_* the Hamiltonian is increasing in B , so its value is at most the right-hand side, whereas the tangent is minimized at $R_{B,0}$.

Along an AI-dominated continuation, the difference between the two sides of (127), divided by Y_t , converges to

$$\alpha(1 - \alpha) \left[1 - \left(\frac{B_*}{\bar{B}} \right)^{(1-\alpha)/\alpha} \right] > 0.$$

To obtain this limit, use $M/Y \rightarrow 0$, $\psi(B)R_B \rightarrow 0$, and $\pi(B_*)/Y \rightarrow \alpha(1 - \alpha)(B_*/\bar{B})^{(1-\alpha)/\alpha}$. Thus the global support condition holds sufficiently late on this continuation even when the stronger curvature condition fails at low counterfactual AI-efficiency levels. The finite transition still requires its own support check.

B.5 Replication files

I use NumPy for array operations and linear algebra (Harris et al., 2020) and Matplotlib to generate the figures (Hunter, 2007). The package versions are recorded in `requirements-rewrite.txt`.

The code, simulation data, and replication instructions are available at <https://github.com/oliverpardo1979/ai-growth-and-labor>. The [replication guide](#) explains how to install the dependencies, check the published results, and recompute the eight trajectories. From the repository root, run:

```
python scripts/simulate_rewrite_illustrative_rsi.py
```

This command solves and audits all four elasticities in both scenarios. Use `--chi 7.5` or `--chi 1.5` for one scenario; `--sigma 1.5`, for example, solves one elasticity. With all four checkpoints available, `--finish` repeats the equilibrium checks and exports the figures. The full replication entry point is `scripts/reproduce_rewrite_results.py`.

The [higher-productivity results folder](#) and [lower-productivity results folder](#) contain the scenario inputs, validated CSV files, outcome summaries, equilibrium checks and figure metadata. Checkpoint and CSV hashes bind the audits to the plotted data. No comparison is exported unless the equilibrium and activation-continuity checks pass for all four elasticities in that comparison.

The repository includes the two comparisons reported here. Earlier experiments are preserved separately and are not needed for replication.

C The uncapped economy with unit elasticity

This appendix derives the rates and allocation ratios supporting Proposition 5 in Section 5.2 and defines the normalized variables used to prove local convergence. I use the uncapped RSI technology (44) and the production function (53). In this appendix and its proofs, a superscript $*$ denotes the reference BGP characterized in Section 5.2, distinguishing it from nearby equilibrium trajectories.

C.1 An explicit balanced-growth equilibrium

At unit elasticity, $s_X = \omega_X$ and $e_X = 1 - (1 - \alpha)\omega_X$. The AI-demand and monopoly conditions, equations (13) and (19), give

$$\frac{p_X X}{Y} = (1 - \alpha)\omega_X, \quad \frac{U}{Y} = (1 - \alpha)^2 \omega_X^2. \quad (128)$$

Substituting $X = BU = (1 - \alpha)^2 \omega_X^2 BY$ into production yields

$$Y^{1-(1-\alpha)\omega_X} = K^\alpha (AL)^{(1-\alpha)\omega_L} [(1 - \alpha)^2 \omega_X^2 B]^{(1-\alpha)\omega_X}. \quad (129)$$

On an exact balanced-growth path with positive constant K/Y , C/Y , and M/Y , log differentiation of (129) and (44) gives the two growth restrictions (54) and (55). Define

$$\Delta \equiv (1 - \alpha)(1 - \eta - \omega_X). \quad (130)$$

For $\Delta > 0$, their unique solution has

$$g_Y^* = \frac{(1 - \eta)(1 - \omega_X)}{1 - \eta - \omega_X} (n + \gamma), \quad (131)$$

and g_B^* as in (60). The equality $g_K = g_C = g_Y^*$ follows from constant capital and consumption ratios; equations (12) and (5) then give (57) and (58).

Define the candidate ratios $k^* = K/Y$, $u^* = U/Y$, $i^* = (\dot{K} + \delta K)/Y$, $m^* = M/Y$, and $c^* = C/Y$ by

$$k^* = \frac{\alpha}{r^* + \delta}, \quad u^* = (1 - \alpha)^2 \omega_X^2, \quad (132)$$

$$i^* = (g_Y^* + \delta)k^*, \quad (133)$$

$$m^* = \frac{u^* \eta g_B^*}{\rho - n + \eta g_Y^*}, \quad (134)$$

$$c^* = 1 - u^* - i^* - m^*. \quad (135)$$

In this appendix, k^* is a capital-output ratio, not capital per unit of effective labor, and i^* is gross physical investment as a share of output. The research share comes from the developer's first-order and costate conditions; the consumption share comes from market clearing. These ratios are endogenous, not fixed saving or research rules. The proof in Appendix A verifies their positivity and constructs the reference initial levels from the production and research equations, for given initial population and labor-augmenting technology. Along the BGP, $B_t = B_0^* e^{g_B^* t} \rightarrow \infty$, $g_K = g_C = g_U = g_M = g_Y^*$, $g_X = g_Y^* + g_B^*$, and $g_q = g_Y^* - g_B^*$.

Since $n + \gamma > 0$, equations (54) and (55) require $\Delta > 0$ for this positive finite-rate BGP. If $\Delta = 0$, they are inconsistent; if $\Delta < 0$, they imply $g_B < 0$, contrary to the RSI technology. Thus $\eta + \omega_X \geq 1$ excludes this BGP, not every possible equilibrium trajectory. The same restriction $\eta + \omega_X < 1$ also suffices for global developer optimality: the proof uses the invertible state transformation $S = B^{(1-\eta)/\eta}$, in which the RSI technology is jointly concave in S, M and operating profit is concave in S . Finally, $\rho > n$ makes both transversality expressions decline and the objective integrals converge.

C.2 Equilibrium trajectories near the balanced-growth path

The BGP supplies a reference for proving existence from nearby initial stocks that need not be on that path. For the given exogenous paths of population and labor-augmenting technology, compare each variable with its value on the corresponding reference BGP at the same date. Define log deviations

$$\xi_K = \log \frac{K}{K^*}, \quad \xi_B = \log \frac{B}{B^*}, \quad \xi_C = \log \frac{C}{C^*}, \quad \xi_q = \log \frac{q}{q^*}. \quad (136)$$

Time subscripts are suppressed: the reference quantities in the denominators follow the BGP rather than remain fixed at their initial values. Substitution into the four equilibrium dynamics gives a smooth autonomous system $\dot{\xi} = f(\xi)$ with $f(0) = 0$, where $\xi = (\xi_K, \xi_B, \xi_C, \xi_q)'$. Let $J^* = Df(0)$ denote its Jacobian. The proof in Appendix A derives its entries directly from equations (29a)–(29d), with $\psi = 1$ and $\psi' = 0$.

The parameter restrictions in Proposition 5 imply two local properties. First, the linearized dynamics have exactly two stable directions and no eigenvalue with zero real part. Writing E^s for the real invariant subspace associated with eigenvalues having negative real parts, the first property is

$$\text{spec}(J^*) \cap i\mathbb{R} = \emptyset, \quad \dim E^s = 2. \quad (137)$$

Second, these stable directions allow independent changes in the two predetermined stocks. Let $V_S \in \mathbb{R}^{4 \times 2}$ be any real basis matrix for E^s , and let V_{KB} consist of its first two rows. Then

$$\det V_{KB} \neq 0. \quad (138)$$

Nonsingularity does not depend on the choice of basis. It allows the stable manifold to select C_0, q_0 for every sufficiently close pair K_0, B_0 , rather than only for stocks on a curve. Both properties follow analytically from the proposition's parameter restrictions; neither is an additional assumption or a calibration-specific test. The complete argument, including both transversality conditions and global agent optimality, is in Appendix A, [Proof of Proposition 5](#).

No constant growth rates or allocation shares are imposed along these transitions. The result is local in initial stocks, not in time. In particular, it concerns equilibrium choices of consumption and research, not convergence from arbitrary nearby values of all four variables: there are also two unstable directions. It does not rule out equilibria that leave the local neighborhood. The normalized Jacobian is independent of χ , A_0 , and N_0 ; changing research productivity changes the reference levels, not the eigenvalues governing local convergence.

C.3 Growth and distribution without an upper bound

The uncapped BGP has

$$\frac{wL}{Y} = (1 - \alpha)\omega_L > 0, \quad \frac{(r + \delta)K}{Y} = \alpha, \quad \frac{\Pi}{Y} = (1 - \alpha)\omega_X - u^* - m^*. \quad (139)$$

Equation (57) shows that wages and output per person grow together at a rate above γ , even though labor retains a positive, constant income share. Without an upper bound, continued growth in AI efficiency sustains this wage-growth premium. With any upper bound, the unit-elastic equilibria in Proposition 1(ii) instead have $g_{Y/N}, g_w \rightarrow \gamma$. The equivalence between wage growth above γ and a vanishing labor share across the regimes with an upper bound does not extend to the uncapped economy.

Within the restrictions of Proposition 5, increasing the AI weight raises the common growth rate and the interest rate:

$$\frac{\partial g_{Y/N}^*}{\partial \omega_X} = \frac{\partial g_w^*}{\partial \omega_X} = \frac{\partial r^*}{\partial \omega_X} = \frac{\eta(1-\eta)(n+\gamma)}{(1-\eta-\omega_X)^2} > 0. \quad (140)$$

Labor’s income share falls across these parameterizations, but remains constant over time on each BGP. As $\omega_X \rightarrow 0^+$, the common per-person growth rate converges to γ and $r^* \rightarrow \rho + \gamma$. These statements compare balanced-growth equilibria; they do not describe a transition from a no-AI economy or establish an uncapped equilibrium for $\sigma > 1$.

Declaration of AI use

During the preparation of this work, I used OpenAI Codex to assist with mathematical derivations, consistency checks, manuscript editing, and the preparation and testing of computational code. I take full responsibility for the content of the paper.

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